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SECP2613-02 SYSTEM ANALYSIS AND DESIGN (WBL)

PROJECT 1 PHASE 3

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1.0 OVERVIEW OF THE PROJECT

Early diagnosis is critical in providing assistance for children with autism. However, many current approaches to autism screening remain inefficient and inaccessible for both families and healthcare workers.

This project proposes a prototype called the Autism Early Screening Support System, AutiScreen. The system aims to provide a technological means to facilitate the screening process, improve accessibility, and deliver accurate, standardized results. To ensure wide availability, the system is developed for both mobile and web platforms.

The system serves three main stakeholder groups, which are the parents/ caregivers, autistic children, and non-profit organization (NGO), which acts as the system owner. Core functionality includes the M-CHAT-R screening module and child profiling based on the BPPOKU Pindaan 2019 (Bhg A) form. Building the AS-IS analysis conducted in Phase 2, this phase presents the TO-BE system design, translating identified gaps into a structured, digital solution for screening, behaviour monitoring, appointment booking, and reporting.

2.0 PROBLEM STATEMENT

Based on the business process analysis and interview findings from Phase 2, the following core problems were identified in the current manual system:

2.1. The Disjointed and Inaccessible of the Process

Current methods of assessing whether a kid is developing normally require excessive resources and accessibility problems. Moreover, the fragmented nature of the process may make it difficult for parents to find out what to do first when suspecting that their child is underdeveloped.

2.2 Overreliance on Manual Screening Practices

Despite the availability of more advanced technologies, traditional screening procedures are still very dependent on manual labor. Due to this fact, as well as other reasons related to the lack of communication and cooperation, screening often becomes delayed.

2.3 Existence of Systemic Obstacles and Absence of Integration

In most cases, there is a low degree of awareness as well as the absence of standardization. In addition, the lack of integration between the health care and education sectors leads to a "silo mentality" situation when the exchange of information by the parties accountable for a child's development is rather poor.

2.4 No Technology-Based Centralization Solution

The lack of centralization, with no technology-based solution, leaves room for inefficiencies in terms of data management as well as coordination of activities by various NGOs and parents. Tracking of results for screening tests, such as M-CHAT-R, and managing profiles based on BPPOKU guidelines will not be efficient without technological solutions.

2.5 Absence of Post-Therapy Support

As confirmed in the Phase 2 interview, once the limited monthly clinical session ends, parents are left without any structured way to track behaviour or flag concerning patterns. This concerns the post-therapy support that needs to be given towards the autistic children.

As a result of such issues, the problem area described above is one wherein a delay in providing support for children with symptoms of autism may occur.

3.0 PROPOSED SOLUTIONS

The recommended solution is known as the Autism Early Screening Support System, which is an advanced two-fold application Web and Mobile technology system. The system combines standard medical screening procedures with daily management systems to create an efficient technological platform for the use of parents, children, and NGOs.

3.1 Core System Infrastructure and Security

The basis of the system lies in data security and privacy.

3.1.1 Authentication and Access Controls

The system uses safe methods for logging into and out of the system, which ensures that confidential data is protected by not allowing access to people who are not authorized to see the data.

3.1.2 Registration and Account Management

A separate module of the system is used to manage users' accounts.

3.1.3 Data Security and Privacy

Considering the confidentiality of data in question, data security is a critical feature of the core infrastructure of the proposed solution, as it will serve as a reliable environment for the formative years intervention program.

3.2 Standardized Screening & Profile Management

This module digitalizes the old screening processes manually performed to guarantee accuracy and fast intervention.

3.2.1 Digital Profile Management (BPPOKU Pindaan 2019)

The program will digitalize Bahagian A (Biodata) from the BPPOKU (Pindaan 2019) form. This process will include the following important pieces of information:

➤ Personal Details

Name, MyKad / MyKid identification, and contact details.

➤ Education & Professional Experience

Education level and parents/guardians' employment.

➤ Details of Next of Kin

The system collects detailed information about relatives close to the person for further communication in case of necessity.

3.2.2 M-CHAT-R Screening Integration

This feature includes a test with 20 questions performed according to the M-CHAT-R standard. The users provide "Yes" or "No" answers regarding children's behavior, such as do they react to a gesture or respond when called?

3.2.3 Automated Scoring Algorithm of Risk Level

The system will be able to provide risk levels automatically, making sure that each child undergoes an evaluation process that is not only consistent but also standardized and objective in nature. For those children with results falling within the Low Risk category, which would include any score of 0-2, there would not be a need to pursue any additional actions unless there are any future worries. Those with scores ranging from 3-7 are considered to have Moderate Risk, automatically prompting notification about the requirement of taking some actions to further examine their behavior. In instances where the scores range from 8-20, the result will be classified as a High Risk automatically, skipping other steps to indicate the urgent requirement of referral of the client for professional diagnostic testing.

3.3 Support, Analytics and Specialized Management

In addition to the diagnosis, this system features modules which enable monitoring and assistance on a continuous basis.

3.3.1 Automated Notification System

To resolve the issue of "delayed referrals" in the conventional approach, this module helps generate automated alerts regarding upcoming screenings or follow-up consultations.

3.3.2 Reporting and Analytics

Data analytics provide an opportunity to understand how the development of a child evolves throughout time, as visualized by parents or NGOs.

3.3.3 Behavior Monitoring

This module monitors the emotional state and mood of the child, enabling caregivers to identify any possible triggers for certain behaviors. These are some of the functions of this module:

- Behavior Log Processor

A real-time function that enables the caregiver to log emotional status, particular behavioral instances, and the corresponding environmental antecedents.

- Trigger Analyzer

A pattern-matching algorithm that scans historical logs to identify correlations between environmental factors that repeat themselves and behavioral disruptions that occur again.

- Mood Trend Evaluator

A computation function that calculates and charts baseline emotional movements over specified timelines, providing parents and NGOs with empirical evidence of structural improvement.

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3.3.1 Appointment Scheduling and Referral Management

This module allows parents to book appointments directly with specialist and referral centres, as well as with education centres, through the system. When a child's behavioural log or screening result shows a concerning trend, parents can make an appointment without needing to rely on a phone call, resolving the ambiguity and delay parents currently face once the limited monthly clinical sessions come to an end. The module checks real-time availability of the specialist or referral centre, records the appointment slot, and automatically issues a referral letter and appointment reminder to the relevant party. This directly replaces the current appointment method and the phone call and manual diary check process, which currently takes 1 to 3 working days and produces no automatic reminder.

3.3.2 Educational Placement and Special Education Management

Through the 3.3.2 module, parents are allowed to search, filter, and view the physical locations of specialized education and early intervention centres within their vicinity. When a child's screening result or behavioral logs indicate a need for early learning support, parents can quickly locate nearby facilities, view available services, and access contact information without any manual search delays. This bridges the gap between clinical diagnosis and educational placement by providing a centralized directory of specialized schools.

3.3.3 Community Groups and Peer Support

An interactive communication hub provides a dedicated discussion platform where parents and caregivers can connect, share experiences, and offer mutual peer support. Users can join localized or topic-specific forums to exchange and navigate educational systems. By centralizing communication within a secure environment, system users can reduce the isolation felt by caregivers and eliminate the scattered nature of relying on external social media channels.

4.0 CURRENT BUSINESS PROCESS / WORKFLOW

4.1 Current business process

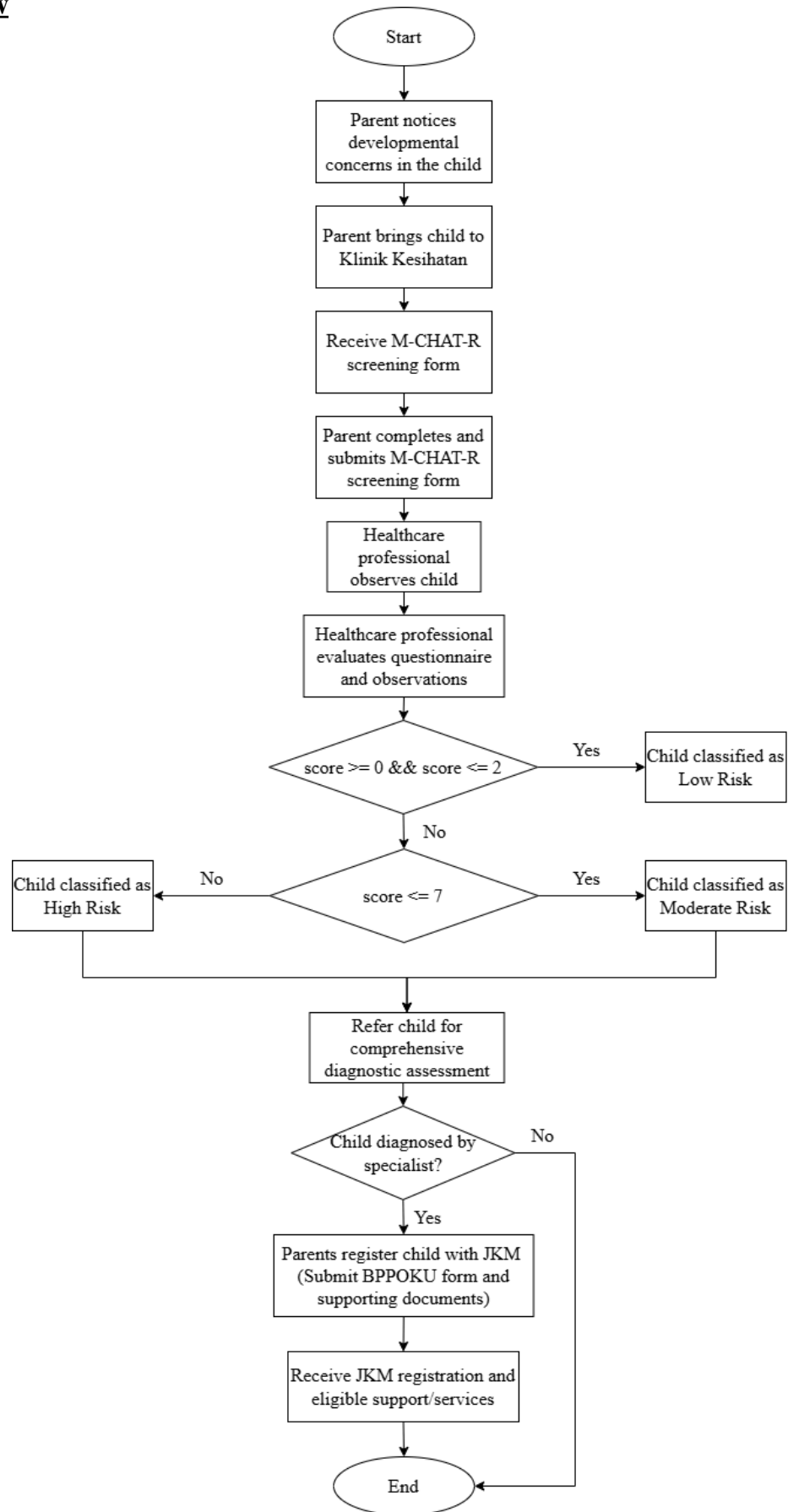
Nowadays, the current system relies heavily on manual processes and traditional methods. The current autism early screening support process is primarily conducted through healthcare professionals, educators or autism specialists. Parents or guardians who suspect developmental delays in their children are required to visit a healthcare centre, clinic or autism intervention centre for an initial assessment.

During the screening session, parents are required to complete paper-based questionnaires which are known as M-CHAT-R regarding the child's communication skills, social interaction, repetitive behaviours, and developmental milestones. Healthcare professionals then observe the child's behaviour and review the completed questionnaire together with the parents.

Based on the collected information, the healthcare professional performs a preliminary screening to determine whether the child shows signs associated with Autism Spectrum Disorder (ASD). If the screening indicates possible autism traits, the child is referred for a comprehensive diagnostic assessment conducted by qualified specialists. If the child is formally diagnosed with Autism Spectrum Disorder (ASD) by a qualified specialist, the parents or guardians may register the child with the Department of Social Welfare (Jabatan Kebajikan Masyarakat, JKM). To complete the registration, they are required to submit the BPPOKU form together with the necessary supporting documents, such as the medical diagnosis report and identification documents. Upon successful registration, the child may become eligible for welfare assistance, support services, and benefits provided by JKM.

The current process has several limitations. The screening process is time-consuming, relies heavily on manual documentation, and requires parents to attend physical appointments. In addition, there is limited accessibility to specialists in certain areas, and parents often experience delays before receiving screening results or recommendations for further assessment.

4.2 Workflow



5.0 LOGIC DFD (AS-IS)

The Data Flow Diagram (DFD) below models the current (AS-IS) manual workflow of the autism screening and record keeping process, based on the business process described in 5.1.

5.1.0 Context Diagram (Level 0)

The Context Diagram shows the AS-IS “Manual Autism Screening & Record Keeping Process” as a one process that depicts the entire system, together with four external entities to exchange information with, which are the Parents/ Caregivers, the Healthcare Worker, the NGO Administrator, and the Specialist/ Referral Center. The Child entity is also included as the healthcare worker directly observes the child’s behavioural patterns during the screening interaction, in addition to information supplied by the Parent entity.

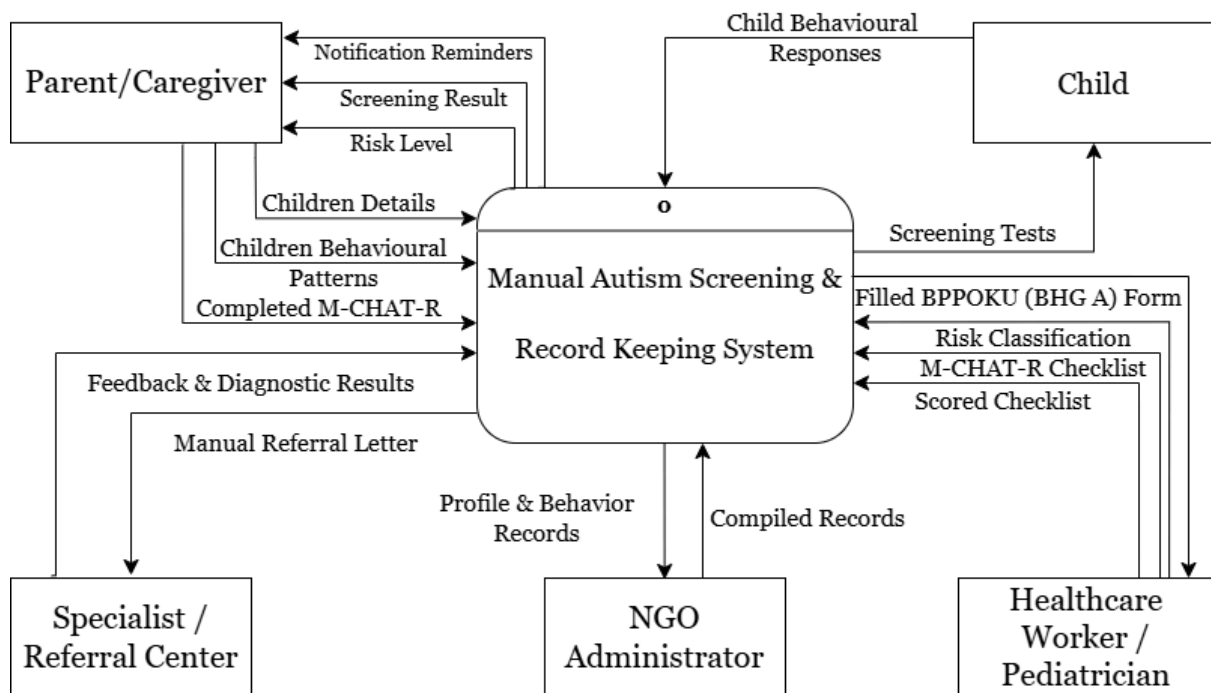


Figure 5.1 Context Diagram (Level 0) AS-IS Manual Autism Screening & Record Keeping System

5.1.1 Diagram 0

The Diagram 0 expands the single AS-IS process into five major manual process and four manual data stores :

- 1.0 Conduct M-CHAT-R Screening: Distribute, complete and manually score the 20 item checklist.
- 2.0 Manage Child Biodata Profile: Collecting and filling the BPPOKU (Bhg A) biodata form

- 3.0 Record Behaviour Observations: Parents manually record behavioural incidents
- 4.0 Schedule Appointments and Referrals: Booking follow-up visits and issuing referrals
- 5.0 Compile Reports: NGO Staff manually compiling screening, behaviour, and appointment records into a report.

The four data stores involved are the D1 Screening Form Files, D2 Child Profile (BPPOKU) Files, D3 Behaviour Log Notebooks, and D4 Appointment Record Book.

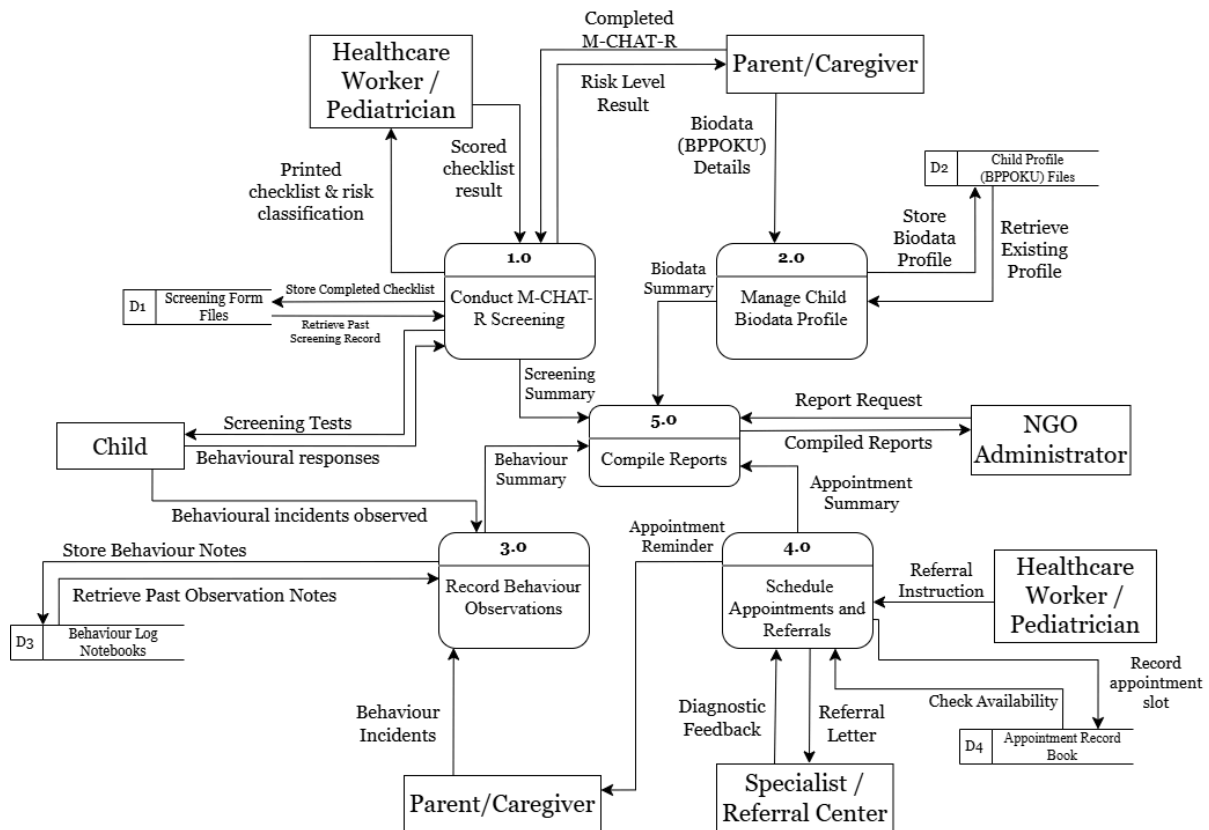


Figure 5.2 Diagram 0 AS-IS Manual Autism Screening & Record Keeping System

5.1.2 Child Diagram: Process 1.0, Conduct M-CHAT-R Screening

Process 1.0 was further expanded into a child diagram as it is the most detailed and critical process in the current manual workflow that involves the 20 question M-CHAT-R checklist, manual scoring, and the risk classification level:

- 1.1 Distribute Screening Checklist: Healthcare workers distribute the checklist form to the parents.
- 1.2 Complete the 20 question Checklist: Parents answer Yes/ No for each of the 20 M-CHAT-R questions based on the child's observed behavioural responses.
- 1.3 Manually Score Checklist: Healthcare Worker tallies the Yes/ No responses by hand.

- 1.4 Determine Risk Level: Healthcare worker classifies the score into Low/ Moderate/ High risk.
- 1.5 Result and Follow-Ups: Healthcare workers inform the parents and issue a manual referral letter to a specialist for High Risk cases.

The four data stores involved are the D1 Screening Form Files, D2 Child Profile (BPPOKU) Files, D3 Behaviour Log Notebooks, and D4 Appointment Record Book.

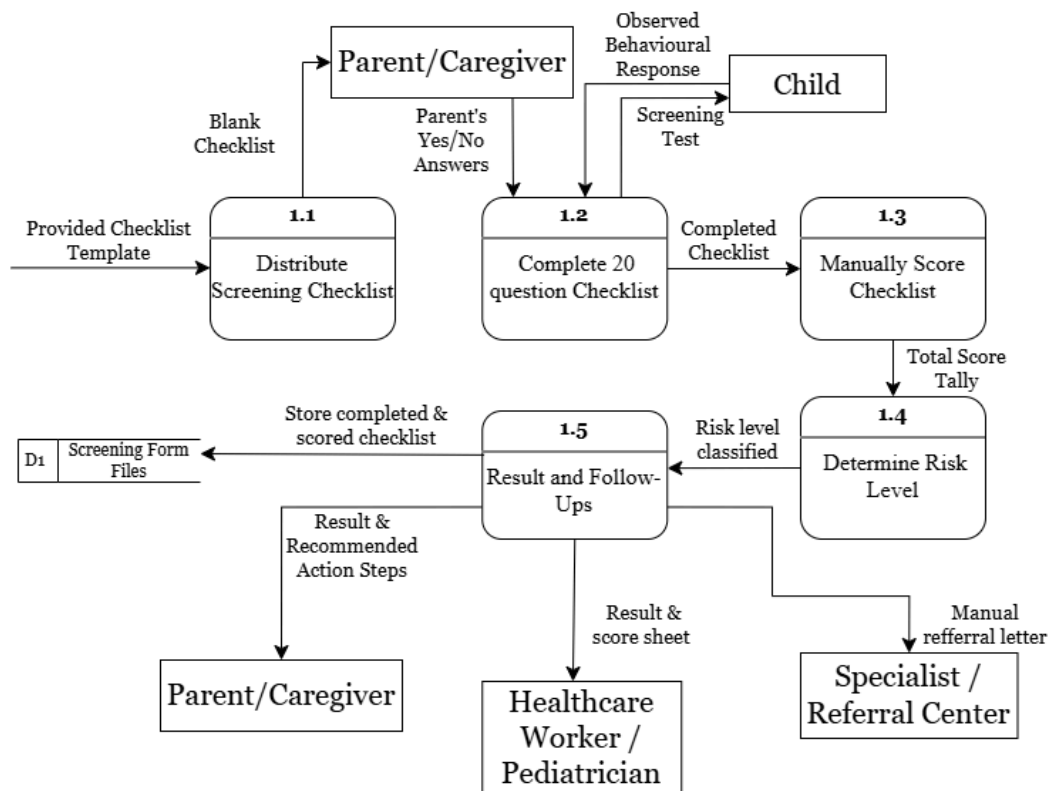


Figure 5.3 Child Diagram (Process 1.0) AS-IS Manual Autism Screening & Record Keeping System

6.0 SYSTEM ANALYSIS AND SPECIFICATION

6.1 Logical DFD TO-BE System

The Logical DFD for the TO-BE system describes the proposed Autism Early Screening Support System. The Logical DFD describes how input data are transformed into outputs used by pediatricians, parents or caregivers, and administrators. The system supports early identification, tracking, and referral of children at risk of autism spectrum disorder (ASD). This representation supports requirement validation, stakeholder communication, and the design and implementation planning.

6.1.1 Context Diagram

The Context Diagram represents the entire system as a single process. It shows all external entities that interact with the system as well as the data flows. The external entities involved are the Parents/ Caregivers, the Child, the NGO Administrator as well as the Specialists/ Referral Center.

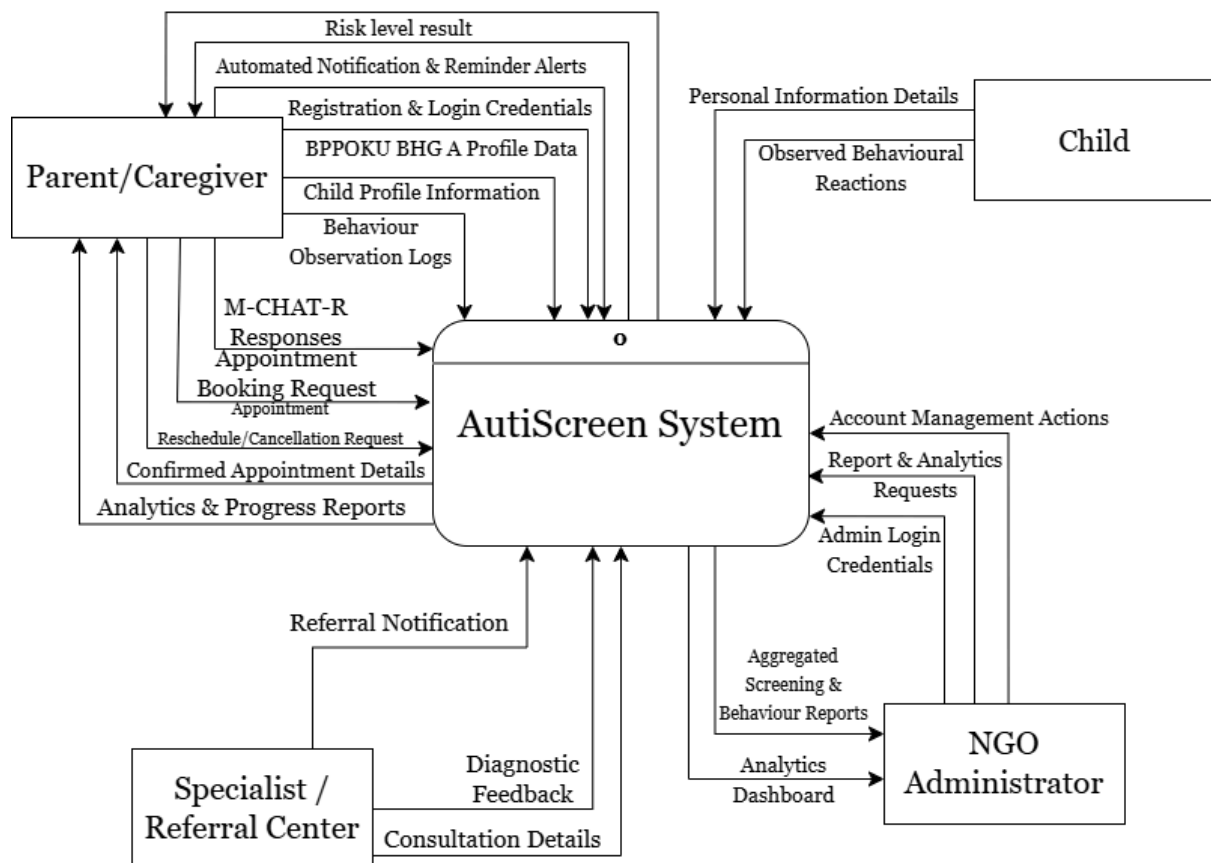


Figure 6.0 Context Diagram TO-BE Automated Autism Screening & Record Keeping System

6.1.2 Diagram 0

Diagram 0 expands the Context Diagram into its major sub-processes and introduces the data stores:

- 1.0 Authentication and Account Management: Create, verify, and manage user accounts that support sign-up, login, password reset, and role assignment for parents and administrators.
- 2.0 BPPOKU Profile Management: Collect, validate, and update the BPPOKU (Bhg A) biodata form and related child profile details while acquiring consent and contact information.
- 3.0 M-CHAT-R Screening: Deliver, complete, and automatically score the 20-item M-CHAT-R checklist while recording responses and highlight the item that requires further follow-up.
- 4.0 Notification and Reminder Management: Generate and send notifications and reminders.
- 5.0 Behavior Monitoring: Capture and log reported behaviour incidents and observational notes with option to upload video or images.
- 6.0 Reporting and Analysis: Gathers screening, behaviour, referral, and appointment data into operational reports and analytics dashboards.
- 7.0 Appointment and Centre Management: Schedule, confirm, and manage appointments and centre resources.

The data stores involved are the D1 User Account Store, D2 Child Profile Store, D3 Screening Results Store, D4 Behaviour Log Store, D5 Appointment Store, D6 Notification Log.

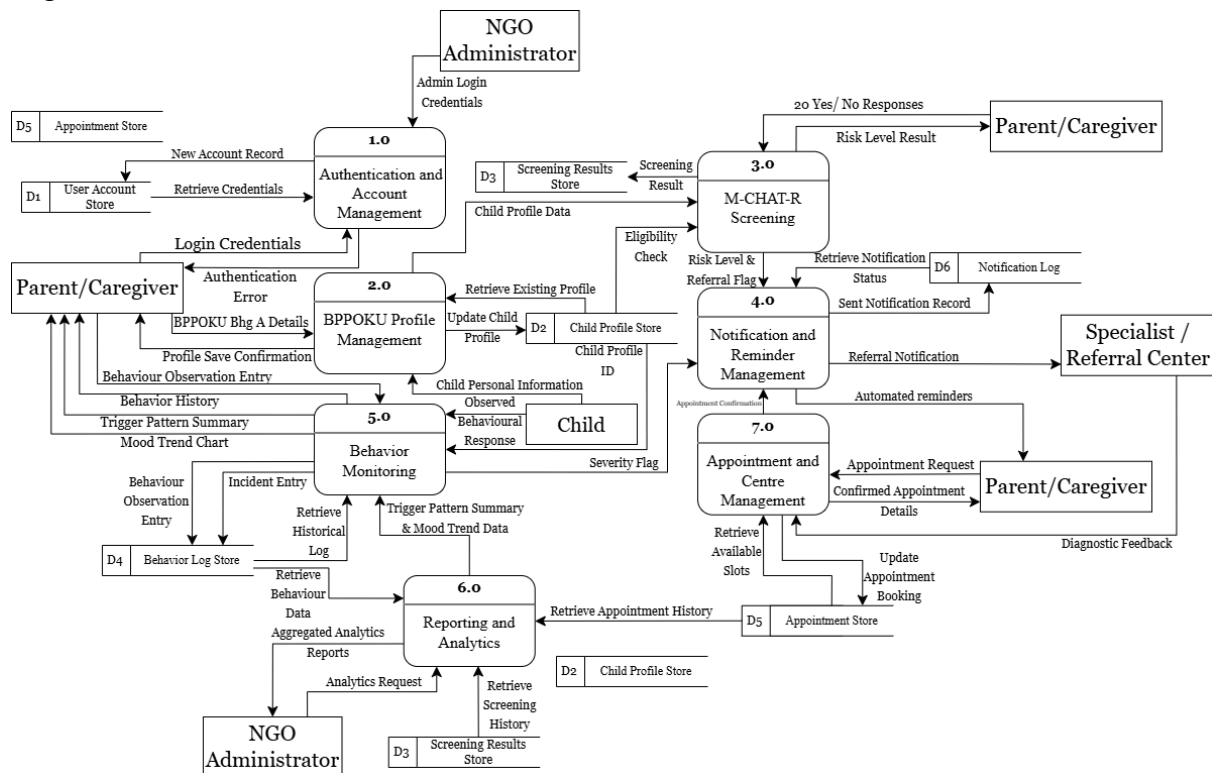


Figure 6.1 Diagram 0 TO-BE Automated Autism Screening & Record Keeping System

6.1.3 Child Diagram

Child diagrams expand Level 1 processes into greater detail. The high-level functions are further decomposed into sub-processes that show the sequence of tasks and the specific data flows between each task.

6.1.3.1 Child Diagram 1.0: Authentication and Account Management

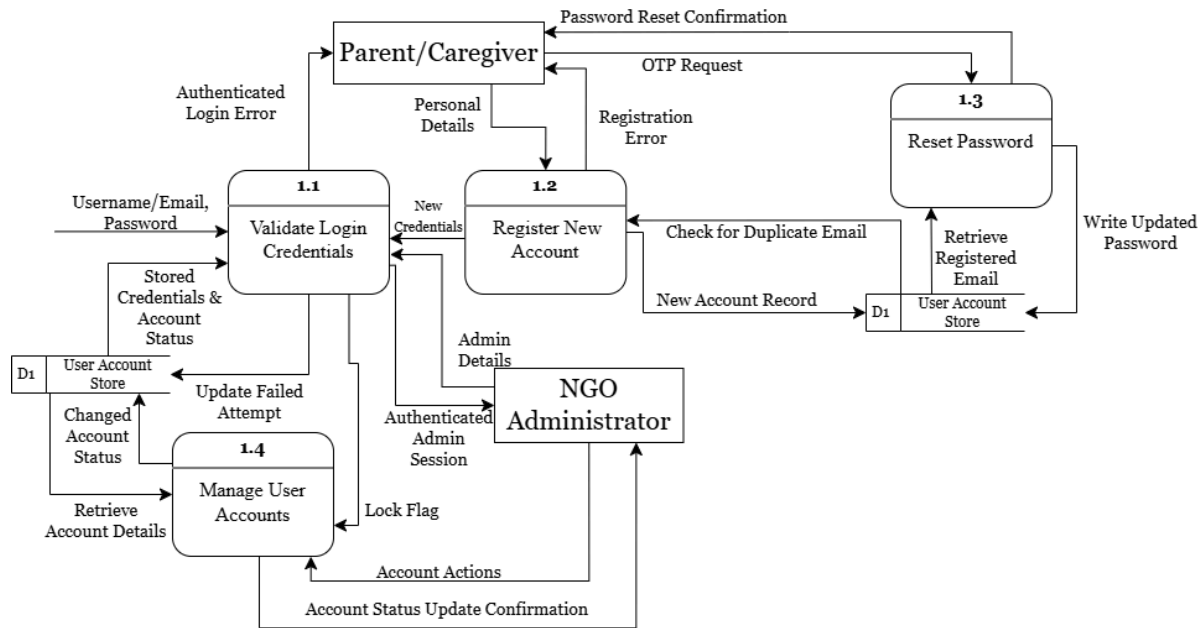


Figure 6.2 Child Diagram (Process 1.0) TO-BE Automated Autism Screening & Record Keeping System

6.1.3.2 Child Diagram 2.0: BPPOKU Profile Management

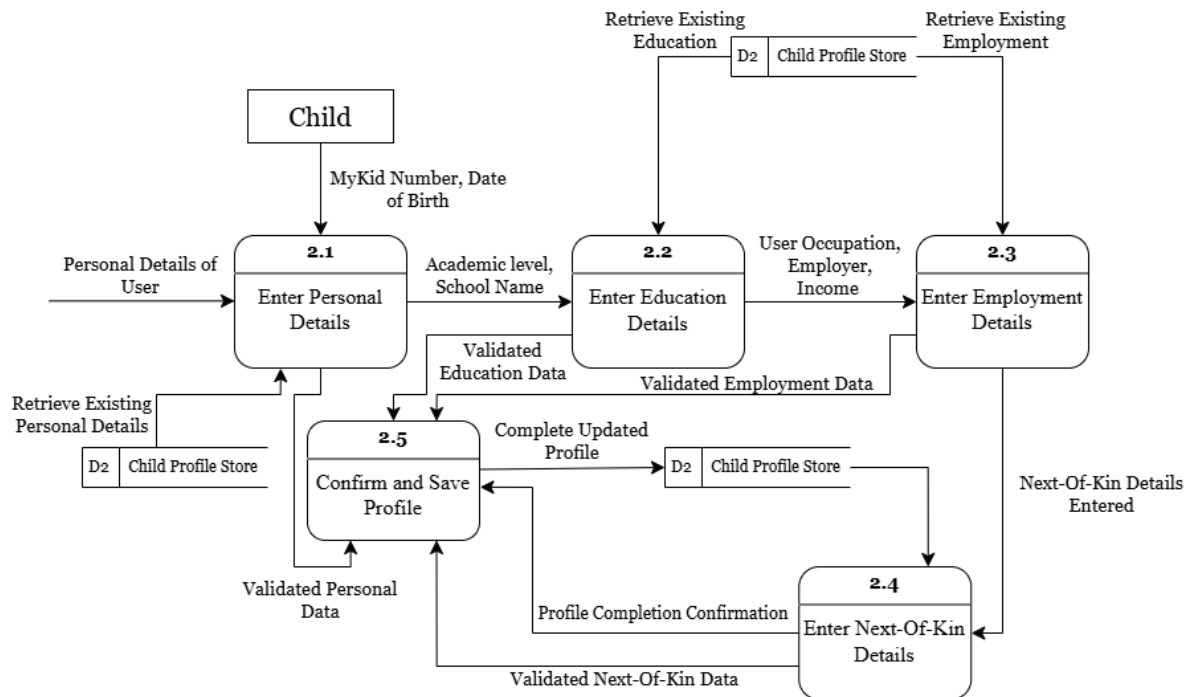


Figure 6.3 Child Diagram (Process 2.0) TO-BE Automated Autism Screening & Record Keeping System

6.1.3.3 Child Diagram 3.0: M-CHAT-R Screening

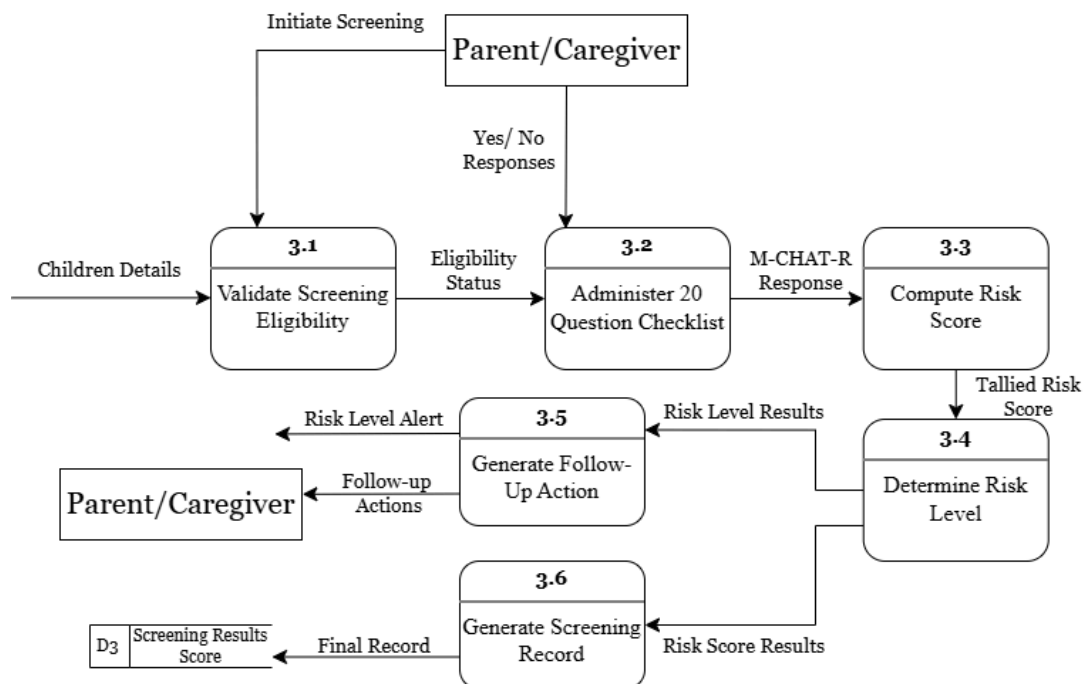


Figure 6.4 Child Diagram (Process 3.0) TO-BE Automated Autism Screening & Record Keeping System

6.1.3.4 Child Diagram 4.0: Notification and Reminder Management

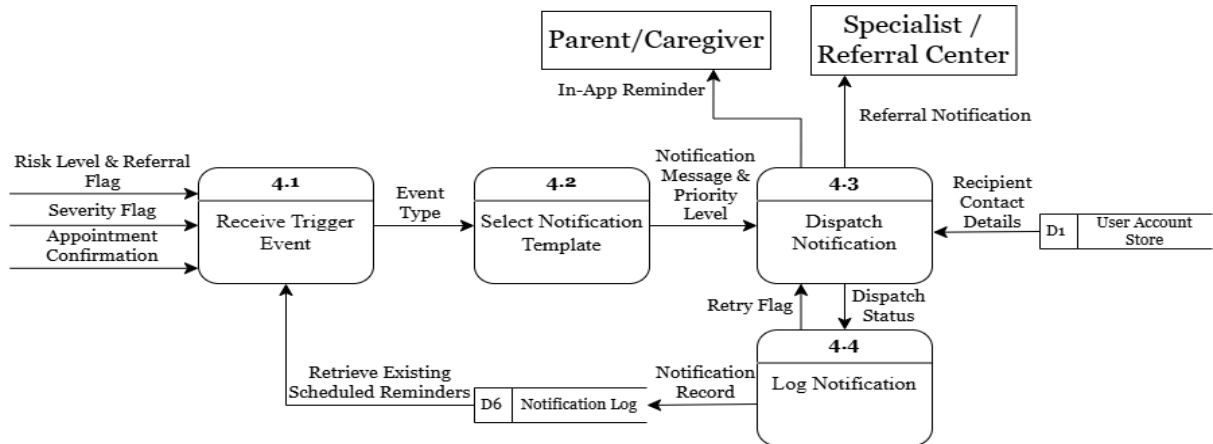


Figure 6.5 Child Diagram (Process 4.0) TO-BE Automated Autism Screening & Record Keeping System

6.1.3.5 Child Diagram 5.0: Behaviour Monitoring

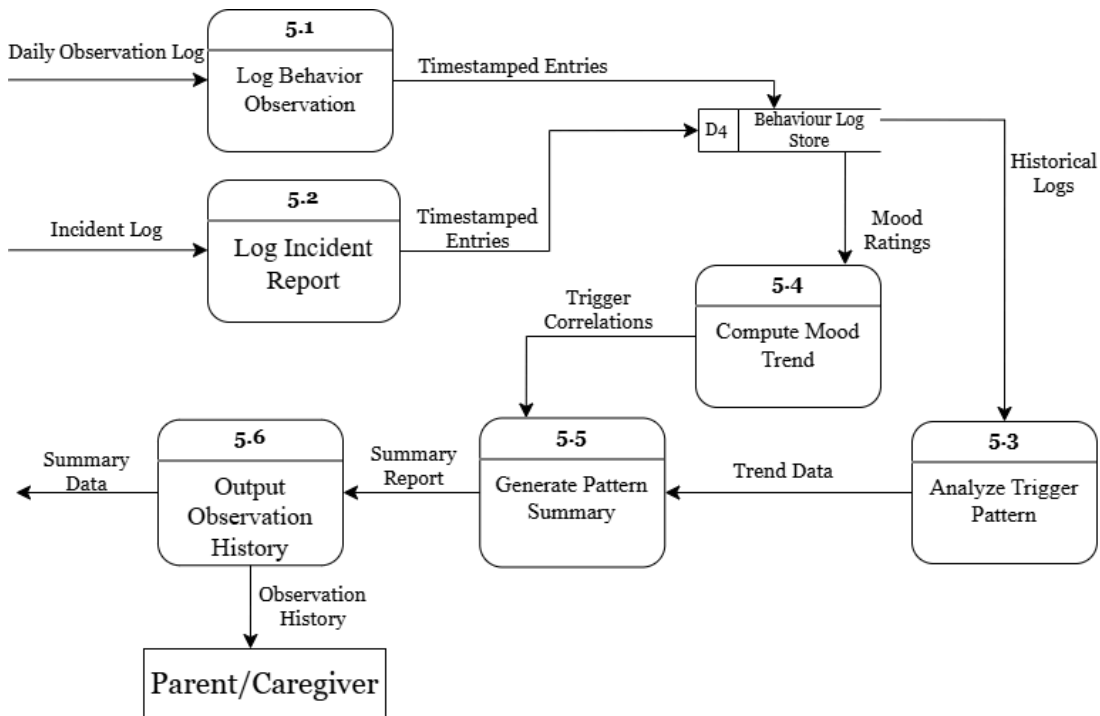


Figure 6.6 Child Diagram (Process 5.0) TO-BE Automated Autism Screening & Record Keeping System

6.1.3.6 Child Diagram 6.0: Reporting and Analytics

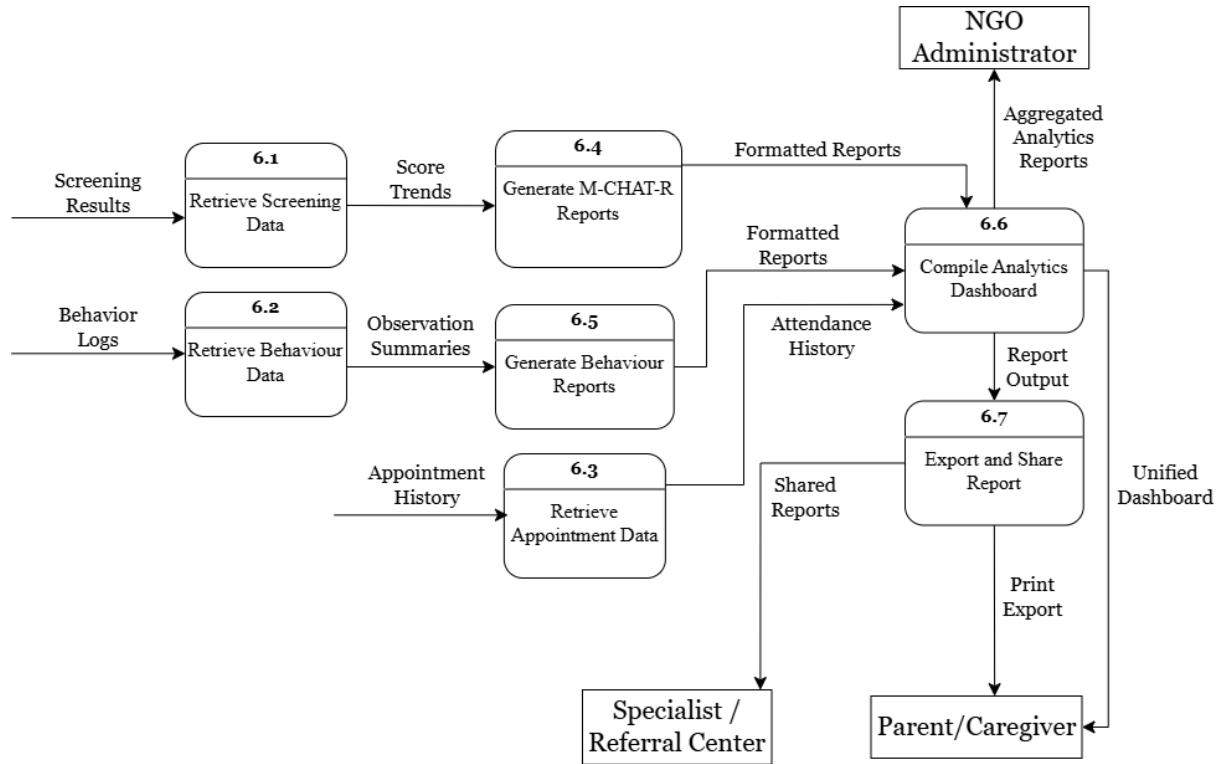


Figure 6.7 Child Diagram (Process 6.0) TO-BE Automated Autism Screening & Record Keeping System

6.1.3.7 Child Diagram 7.0: Appointment and Centre Management

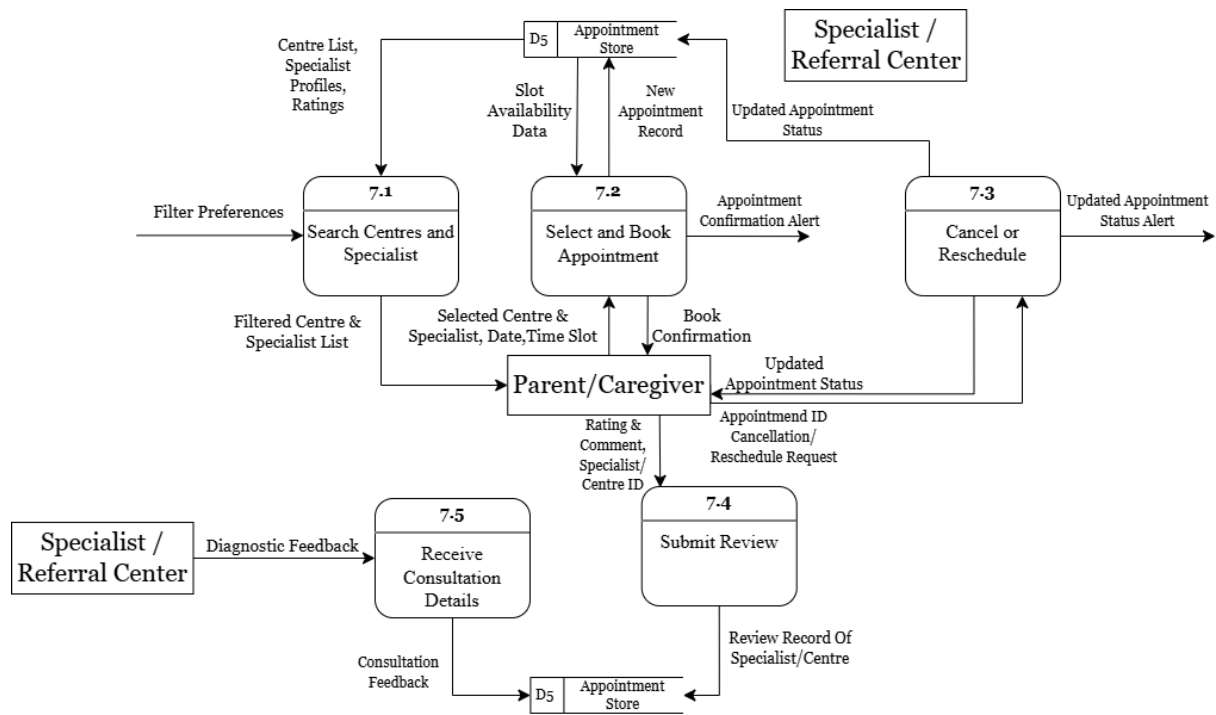


Figure 6.8 Child Diagram (Process 7.0) TO-BE Automated Autism Screening & Record Keeping System

6.2 Process Specification

6.2.1 Process 1.0

Process Specifications: 1.0 Authentication and Account Management	
Process ID	1.0
Process Name	Authentication and Account Management
Process Description	Manages secure login, logout, account registration, and password management for all system users, which are the Parents and NGO Administrators.
Input Data Flows	Login Credentials (Username/Email, Password) Registration Data (Name, Role, Contact Details) OTP Code New Password
Output Data Flows	Authenticated Session Token

	Account Record stored in D1 User Account Store Authentication Error Message OTP Request to Parent/Caregiver Password Reset Confirmation
Type of Process	Online
Sub-processes/ Function	1.1 Validate Login Credentials 1.2 Register New Account 1.3 Reset Password 1.4 Manage User Accounts
Process Logic	1.1 Receive login credentials from the user. 1.2 Validate credentials against D1 User Account Store. IF credentials match THEN GENERATE a session token ROUTE the user to the appropriate dashboard ELSE IF credential do not match THEN RETURN error message INCREMENT failed attempt counter ENDIF IF failed attempts is greater or equals to 3 THEN TEMPORARILY lock account SEND alert to registered email ENDIF 1.3 For new registration,validate all required fields and check for duplicate email in D1. IF no duplicate THEN ENCRYPT password WRITE new account record to D1 SEND welcome notification. ELSE IF duplicate found REJECT registration RETURN error ENDIF 1.4 For password reset, send OTP to registered phone or email, validate OTP, accept new password and update D1.
Logic Method Reference	Structured English
Unresolved Issues	Lockout duration after 3 failed attempts is not yet defined. OTP delivery channel and validity period have not been finalized.

6.2.2 Process 2.0

Process Specifications: 2.0 BPPOKU Profile Management	
Process ID	2.0
Process Name	BPPOKU Profile Management
Process Description	Digitalizes Bahagian A of the BPPOKU form by collecting and storing comprehensive child and parent or caregiver profile data across four sub-sections, which are the personal details, education details, employment details and next-of-kin details.
Input Data Flows	Personal Details (Name, MyKad, Date of Birth, Address, Contact) Education Details (Academic Level, School Name) Employment Details (Occupation, Employer, Income) Next-of-Kin (Name, Relationship, Contact)
Output Data Flows	Saved Child Profile Record in D2 Child Profile Store Profile Completion Confirmation to Parent Profile data to Process 3.0 and Process 6.0
Type of Process	Online
Sub-processes/ Function	2.1 Enter Personal Details 2.2 Enter Education Details 2.3 Enter Employment Details 2.4 Enter Next-of-Kin Details 2.5 Confirm and Save Profile
Process Logic	2.1 Retrieve existing profile from D2 and pre-populate form fields. 2.2 Validate required fields for each sub-section. IF mandatory fields are empty THEN FLAG missing fields BLOCK save DISPLAY validation message ENDIF 2.3 Validate MyKad format. IF format invalid RETURN field-level error, ENDIF 2.4 On valid submission, write or update profile record in D2. 2.5 Return save confirmation to the Parent dashboard.

	2.6 Make child age available to Process 3.0 for eligibility check.
Logic Method Reference	Structured English
Unresolved Issues	Validation rule for MyKad format needs to be confirmed against the 12 digit MyKad numbering standard. It is unclear whether parents can save a partially completed profile or must complete all four sub-sections before the record is written to D2.

6.2.3 Process 3.0

Process Specifications: 3.0 M-CHAT-R Screening	
Process ID	3.0
Process Name	M-CHAT-R Screening
Process Description	Delivers the 20-item M-CHAT-R checklist to eligible children, captures parent responses, automatically computes and classifies the risk, and generates follow-up actions and a screening record.
Input Data Flows	Children Details Initiate Screening Yes/No Responses (20 Items)
Output Data Flows	Eligibility Status M-CHAT-R Response Risk Level Alert to Parent/Caregiver Follow-up Actions to Parent/Caregiver Screening Record to D3 Screening Results Store
Type of Process	Online
Sub-processes/ Function	3.1 Validate Screening Eligibility 3.2 Administer 20 Question Checklist 3.3 Compute Risk Score 3.4 Determine Risk Level 3.5 Generate Follow-Up Action 3.6 Generate Screening Record
Process Logic	3.1 Retrieve child's age from D2 Child Profile Store and

	validate screening eligibility. IF not eligible THEN RETURN ineligibility message ENDIF 3.2 Administer the 20 question checklist and record each Yes/No response. 3.3 Compute the risk score by tallying “At Risk” responses according to the M-CHAT-R scoring point. 3.4 Determine risk level. IF score 0-2 THEN Risk = Low ELSE IF score 3-7 THEN Risk = Moderate ELSE IF score 8-20 THEN Risk = High ENDIF 3.5 Generate follow-up action and risk level alert based on the determined risk level. 3.6 Write the final screening record to the D3 Screening Results Store
Logic Method Reference	Structured English Decision Table (Risk Level Classification)
Unresolved Issues	Exact eligible age range for the M-CHAT-R needs to be confirmed and validated against the official M-CHAT-R clinical guideline. The minimum re-screening interval before a parent may retake the checklist for the same child has not been defined.

6.2.4 Process 4.0

Process Specifications: 4.0 Notification and Reminder Management	
Process ID	4.0
Process Name	Notification and Reminder Management
Process Description	Generates and sends automated notifications and reminders to Parents based on system triggered events such as risk results, behaviour severity flags, and appointment confirmations.
Input Data Flows	Risk Level and Referral Flag Severity Flag Appointment Confirmation

	Recipient Contact Details Retry Flag
Output Data Flows	In-App Reminder to Parent/Caregiver Referral Notification to Specialist/ Referral Center Notification Record to D6 Notification Log Dispatch Status
Type of Process	Online
Sub-processes/ Function	4.1 Receive Trigger Event 4.2 Select Notification Template 4.3 Dispatch Notification 4.4 Log Notification
Process Logic	4.1 Receive trigger event and identify event type. 4.2 Select the appropriate notification template and priority level based on the event type. 4.3 Retrieve recipient contact details from D1 and dispatch the notification message. IF dispatch fails THEN SET retry flag TRIGGER re-dispatch ENDIF 4.4 Log the notification record to D6 Notification Log 4.5 Retrieve existing scheduled reminders from D6 to avoid duplicate or conflicting alerts.
Logic Method Reference	Structured English
Unresolved Issues	Maximum number of retry attempts and the retry interval after a failed dispatch have not been specified. It is unclear whether parents can fully disable critical alerts or only adjust frequency for non-critical reminders.

6.2.5 Process 5.0

Process Specifications: 5.0 Behaviour Monitoring	
Process ID	5.0
Process Name	Behaviour Monitoring

Process Description	Captures and logs routine behaviour observations and acute incident reports, analyzes trigger patterns and mood trends, and generates summarized observation history for parents.
Input Data Flows	Daily Observation Log Incident Log Historical Log
Output Data Flows	Timestamped Entries to D4 Behaviour Log Store Trigger Correlations Mood Ratings Summary Report Observation History to Parent/ Caregiver
Type of Process	Online
Sub-processes/ Function	5.1 Log Behaviour Observation 5.2 Log Incident Report 5.3 Analyze Trigger Pattern 5.4 Compute Mood Trend 5.5 Generate Pattern Summary 5.6 Output Observation History
Process Logic	5.1 Receive and timestamp daily behaviour observation entries, write to D4. 5.2 Receive and timestamp incident report entries, write to D4. 5.3 Analyze trigger patterns across historical logs retrieved from D4. 5.4 Compute mood trends based on logged mood ratings over the selected period. 5.5 Generate pattern summary combining trigger correlations and trend data. IF a behaviour repeats 2 or more times within the period THEN FLAG as “Repeated Pattern” ENDIF 5.6 Output the observation history and summary data to the Parent/ Caregiver.
Logic Method Reference	Structured English
Unresolved Issues	The exact threshold and time window used to qualify a behaviour as a “Repeated Pattern” have not been finalized. Maximum file size/ format and storage handling for uploaded

	video or image evidence still need to be determined.
--	--

6.2.6 Process 6.0

Process Specifications: 6.0 Reporting and Analytics	
Process ID	6.0
Process Name	Reporting and Analytics
Process Description	Aggregates screening, behaviour, and appointment data into formatted reports and a unified analytics dashboard for parents, specialists, and NGO administrators.
Input Data Flows	Screening Results from D3 Screening Results Store Behaviour Logs from D4 Behaviour Log Store Appointment History from D5 Appointment Store
Output Data Flows	Formatted M-CHAT-R Reports Formatted Behaviour Reports Aggregated Analytics Reports to NGO Administrator Unified Dashboard Shared Reports to Specialist/ Referral Center Report Output to Parent/ Caregiver
Type of Process	Online
Sub-processes/ Function	6.1 Retrieve Screening Data 6.2 Retrieve Behaviour Data 6.3 Retrieve Appointment Data 6.4 Generate M-CHAT-R Reports 6.5 Generate Behaviour Reports 6.6 Compile Analytics Dashboard 6.7 Export and Share Report
Process Logic	6.1 Retrieve screening data and score trends from D3. 6.2 Retrieve behaviour data and observation summaries from D4. 6.3 Retrieve appointment history and attendance data from D5. 6.4 Generate M-CHAT-R report from score trends. 6.5 Generate behaviour report from observation summaries. 6.6 Compile all formatted reports into the unified analytics

	dashboard. 6.7 On request, export and share the report. IF recipient equals to NGO Administrator THEN SEND aggregated analytics report ELSE IF recipient equals to Parent THEN ENABLE print/ export ELSE IF recipient equals to Specialist THEN SHARE report ENDIF
Logic Method Reference	Structured English
Unresolved Issues	The frequency of analytics dashboard refreshes has not been decided. The exact permission boundaries for NGO Administrators viewing aggregated and individual child level data still need to be confirmed for privacy compliance.

6.2.7 Process 7.0

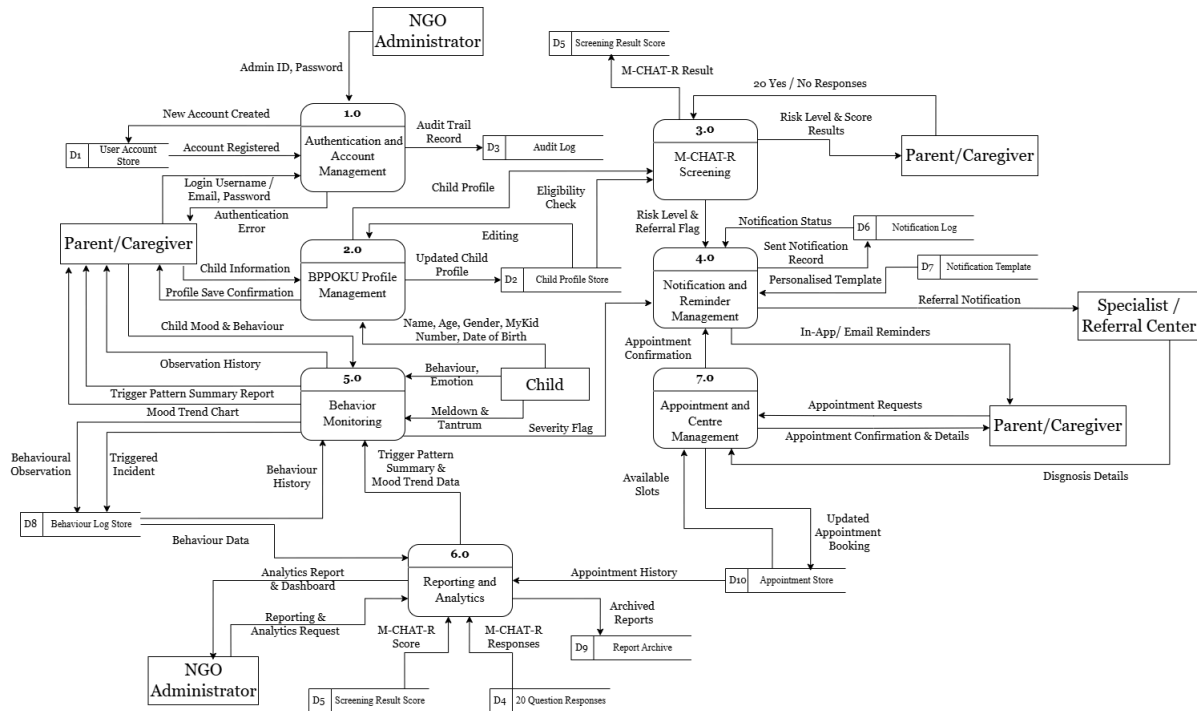
Process Specifications: 7.0 Appointment and Centre Management	
Process ID	7.0
Process Name	Appointment and Centre Management
Process Description	Handles the full cycle of appointment booking, viewing, rescheduling, and cancellation while including centre and specialist search.
Input Data Flows	Filter Preferences Selected Centre and Specialist, Date/ Time Slot Appointment ID, Cancellation/ Reschedule Request Rating and Comment, Specialist/ Centre ID Diagnostic Feedback, Consultation Details
Output Data Flows	Filtered Centre and Specialist List Book Confirmation Appointment Confirmation Alert Updated Appointment Status Alert New Appointment Record to D5 Appointment Store Review Record to D5 Appointment Store

Type of Process	Online
Sub-processes/ Function	7.1 Search Centres and Specialist 7.2 Select and Book Appointment 7.3 Cancel or Reschedule 7.4 Submit Review 7.5 Receive Consultation Details
Process Logic	7.1 Search centres and specialists from D5 based on filter preferences and return filtered list. 7.2 Select and book appointment. IF selected slot is available THEN WRITE new appointment record to D5 RETURN booking confirmation ELSE RETURN slot unavailable error ENDIF 7.3 For reschedule/ cancellation requests, validate appointment ID against D5. IF valid THEN UPDATE appointment status RETURN updated status alert ENDIF 7.4 On consultation completion, receive rating/ comment and submit review, write review record to D5. 7.5 Receive consultation details and diagnostic feedback from Specialist/ Referral Center for the corresponding appointment.
Logic Method Reference	Structured English
Unresolved Issues	Cancellation/ reschedule policy has not been defined. It is unclear whether reviews are moderated before being published or visible to other parents immediately after submission.

7.0 PHYSICAL SYSTEM DESIGN

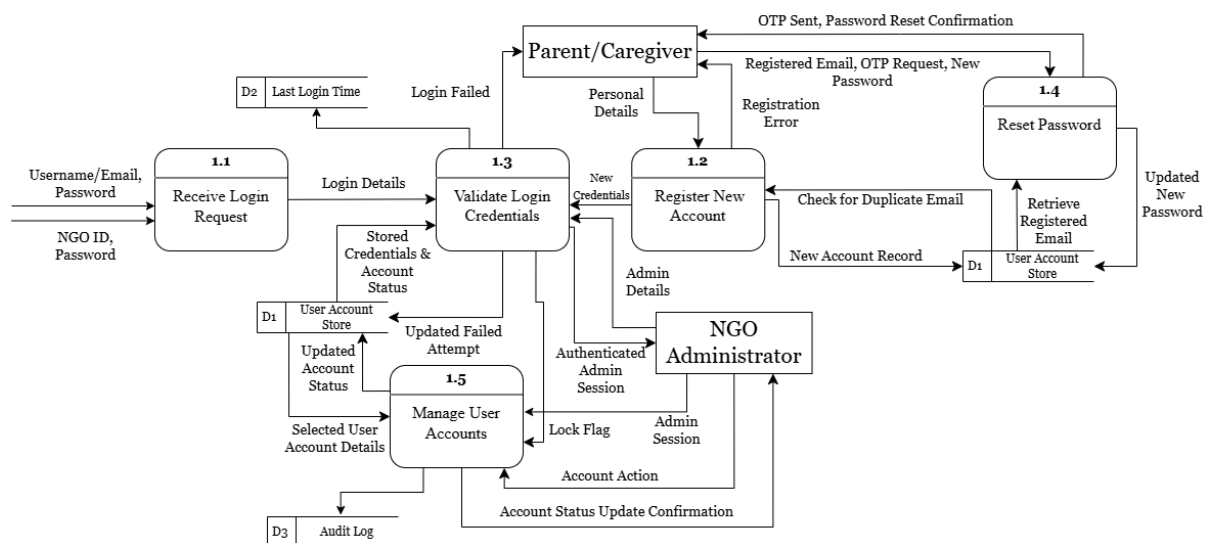
7.1 Physical DFD TO-BE System

7.1.1 Diagram 0

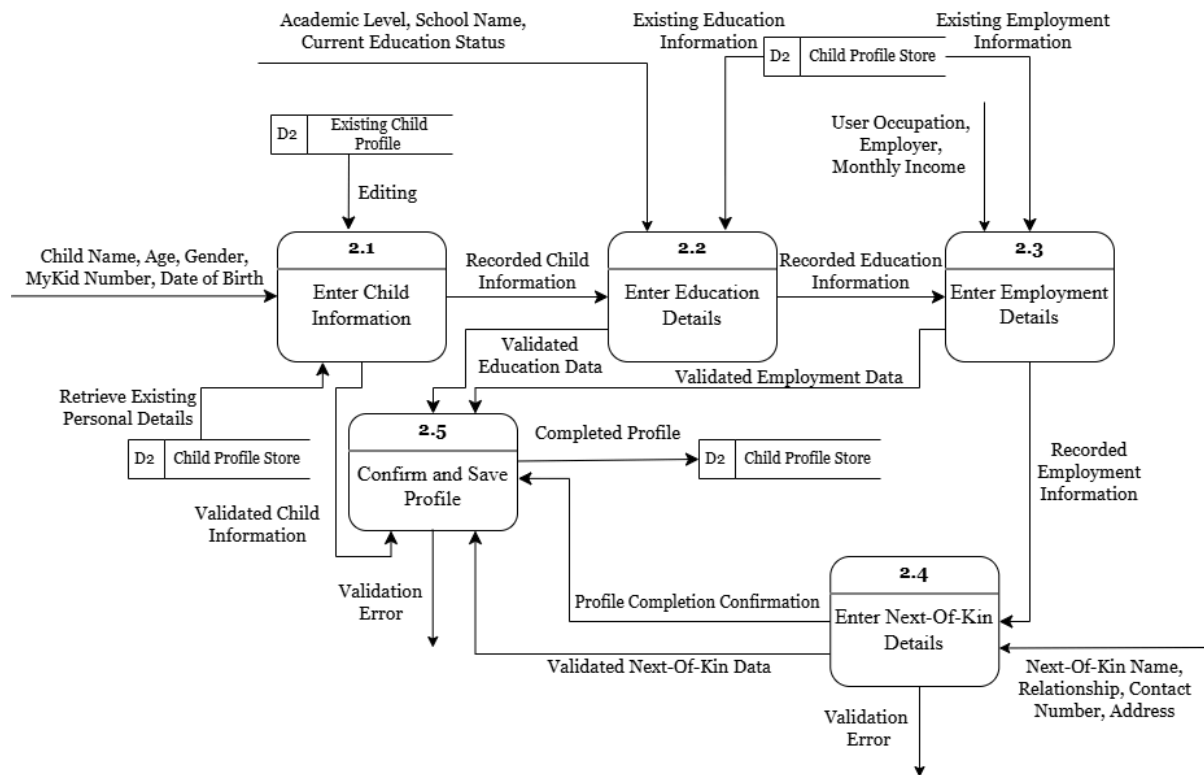


7.1.2 Child Diagram

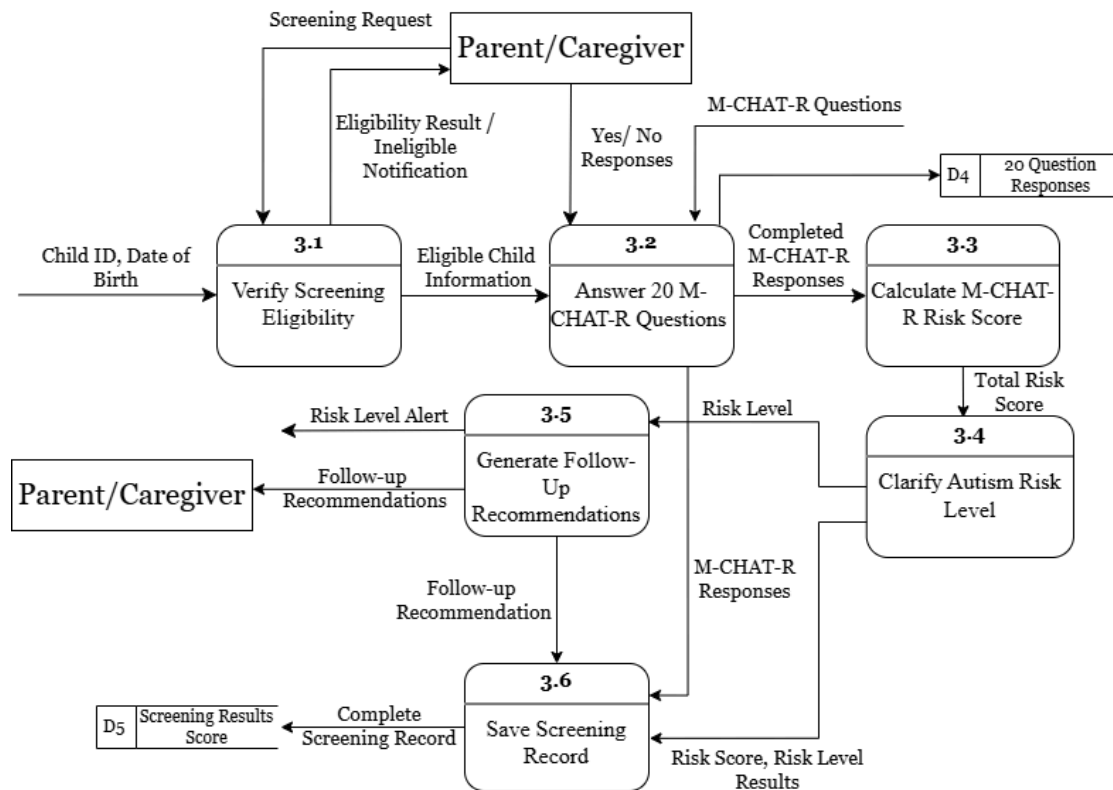
7.1.2.1 Child Diagram 1.0: Authentication and Account Management



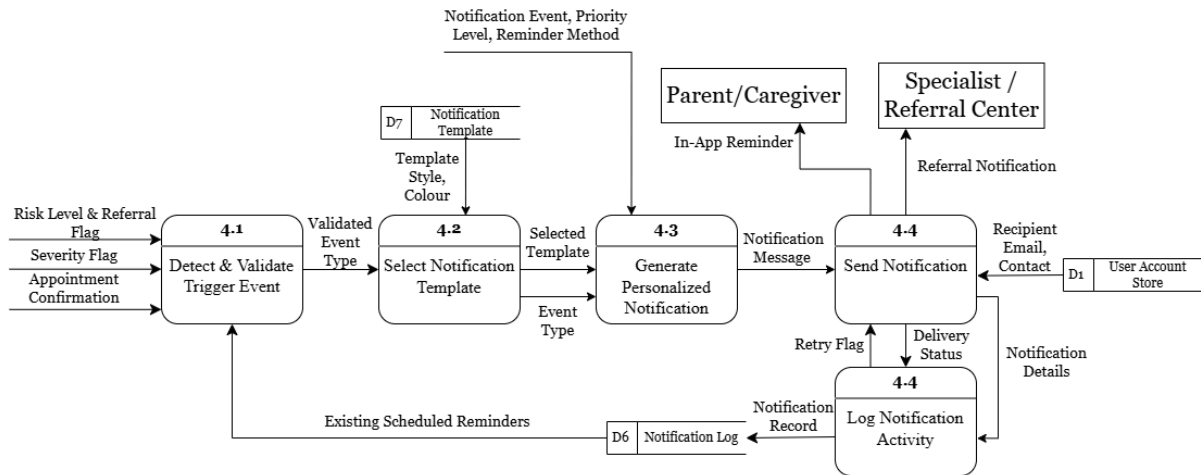
7.1.2.2 Child Diagram 2.0: BPPOKU Profile Management



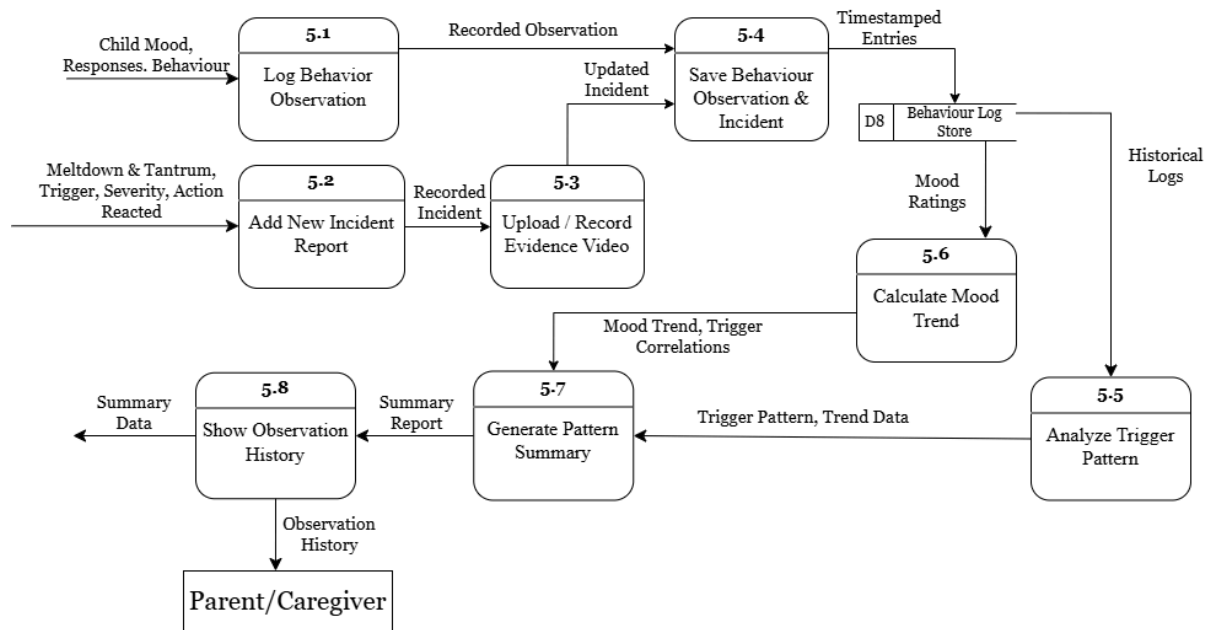
7.1.2.3 Child Diagram 3.0: M-CHAT-R Screening



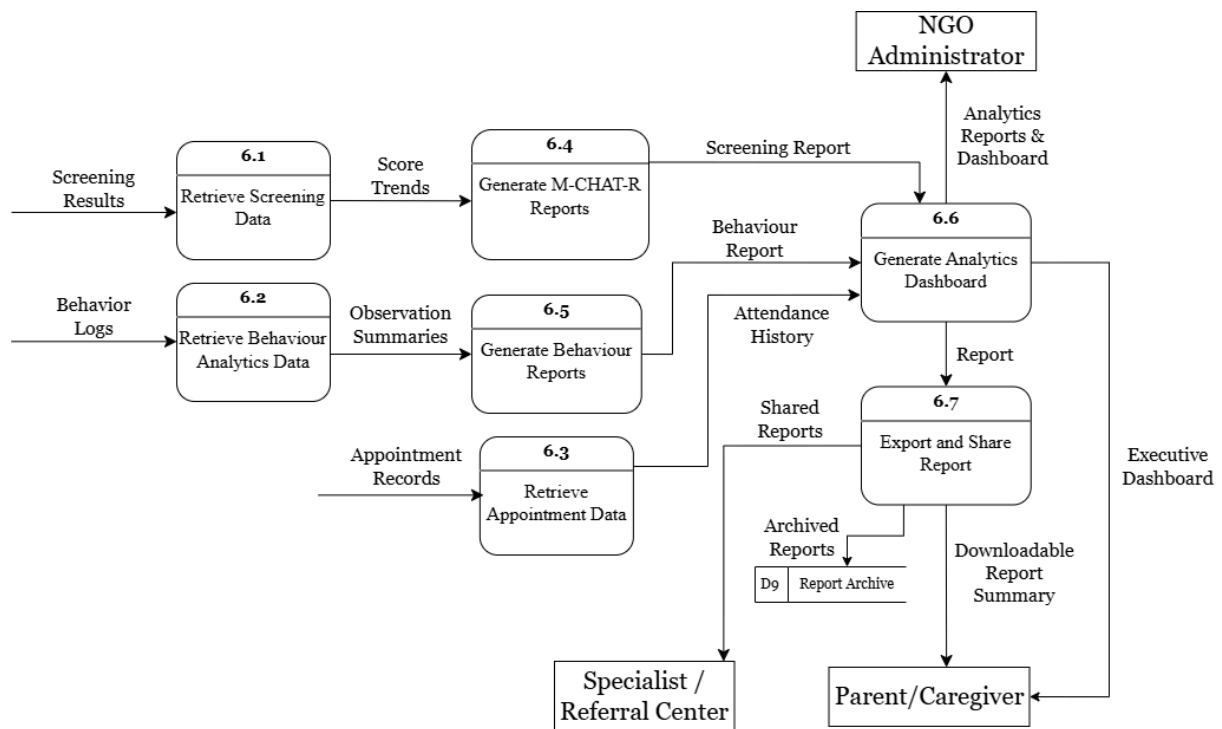
7.1.2.4 Child Diagram 4.0: Notification and Reminder Management



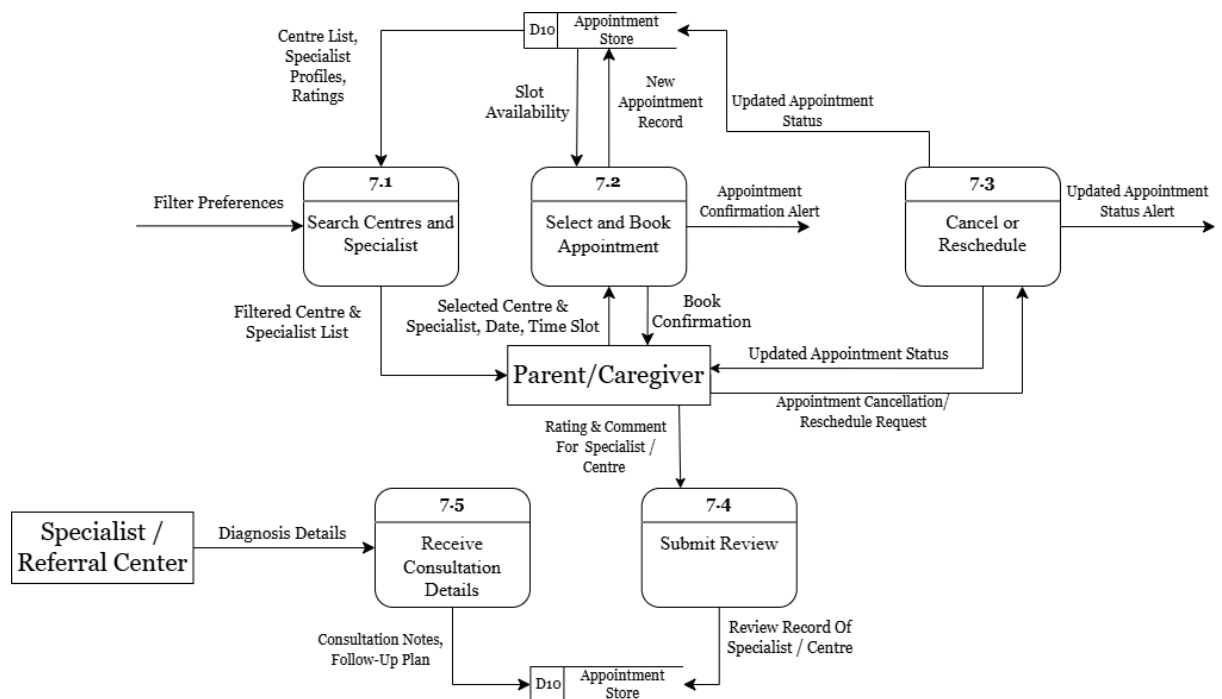
7.1.2.5 Child Diagram 5.0: Behaviour Monitoring



7.1.2.6 Child Diagram 6.0: Reporting and Analytics

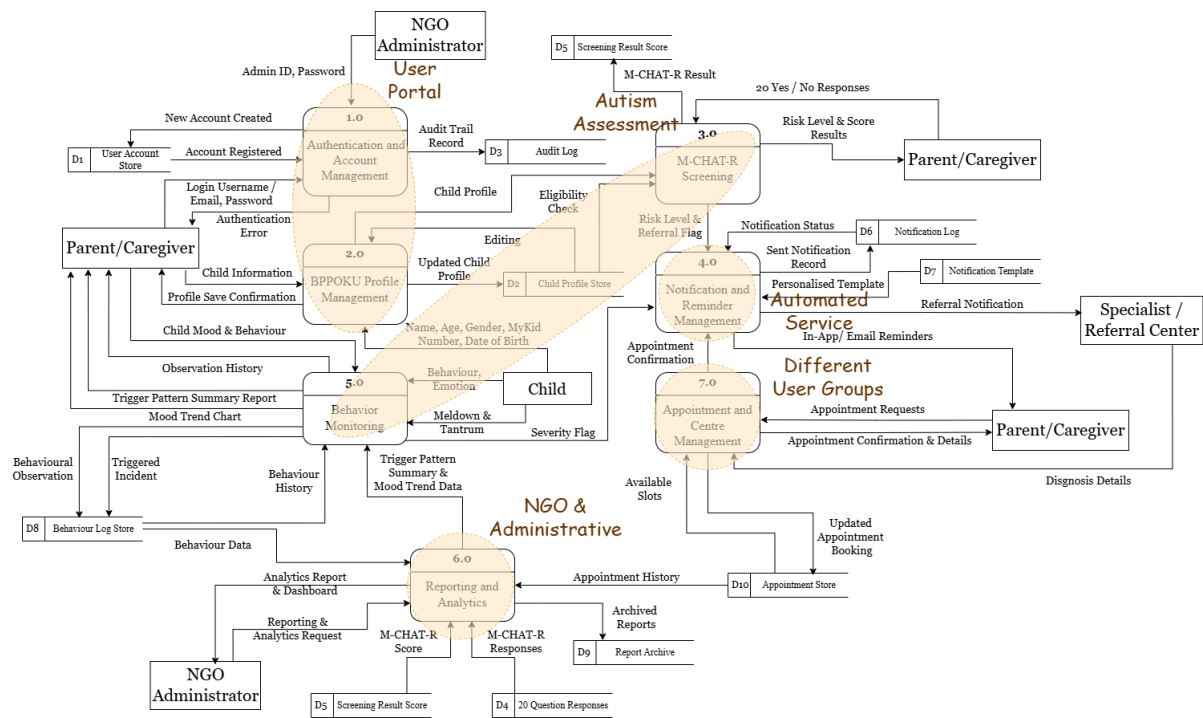


7.1.2.7 Child Diagram 7.0: Appointment and Centre Management



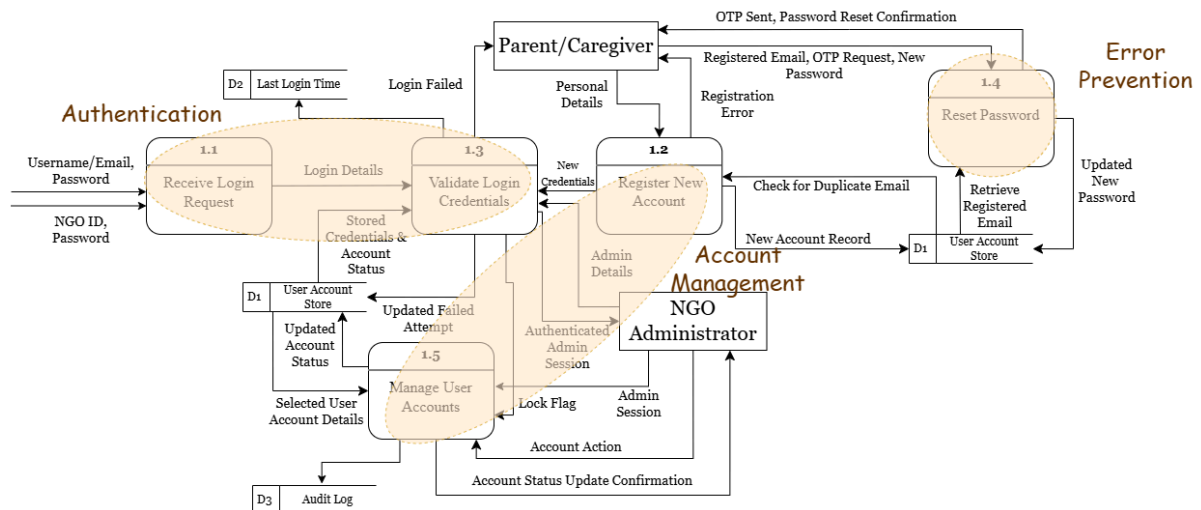
7.1.3 Partitioning

7.1.3.1 Diagram 0

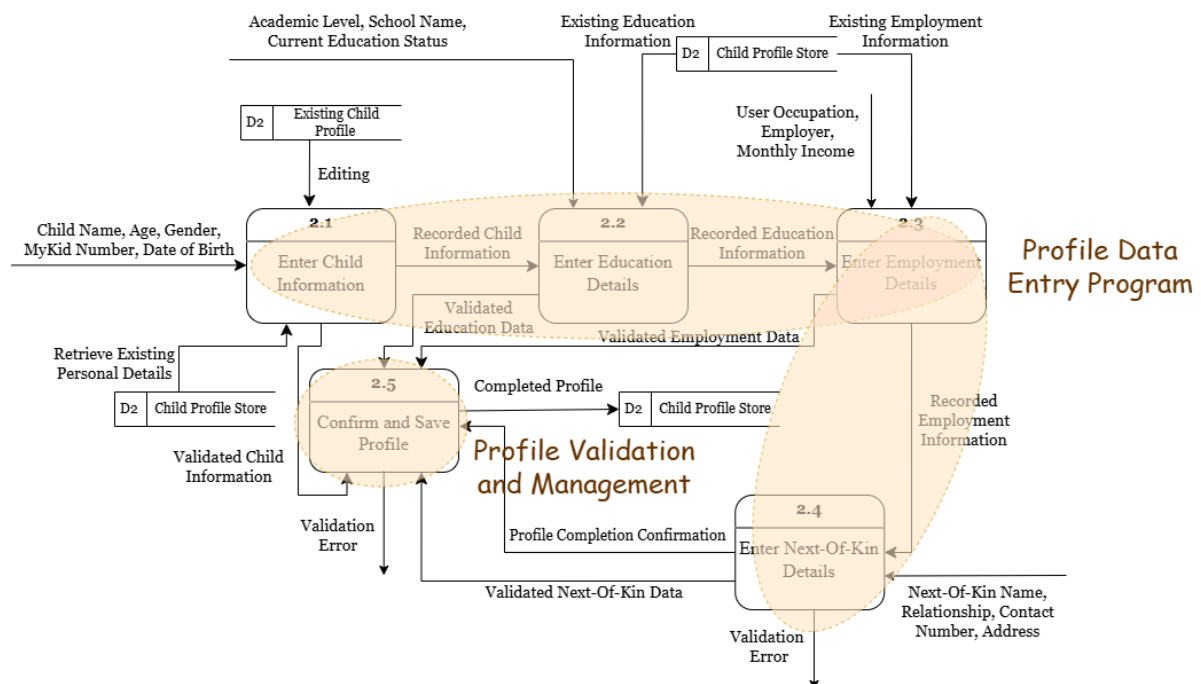


7.1.3.2 Child Diagram

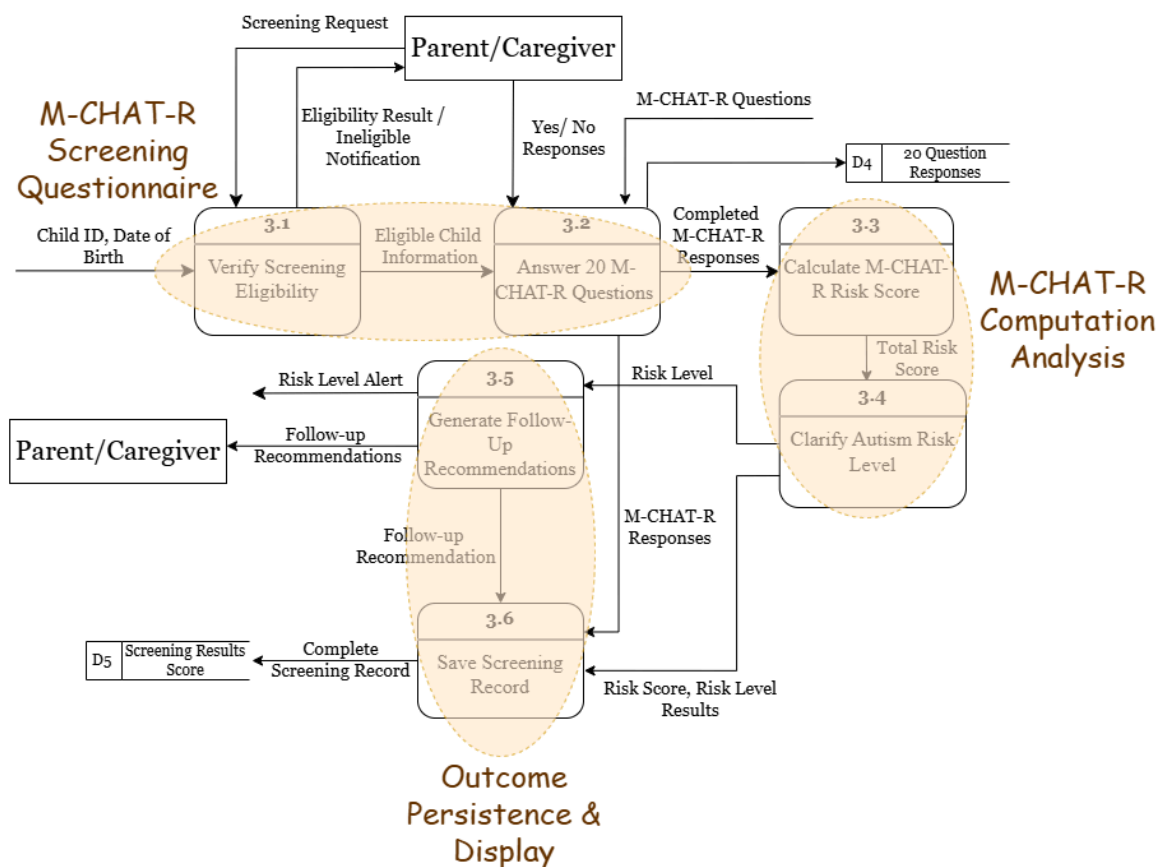
7.1.3.2.1 Child Diagram 1.0: Authentication and Account Management



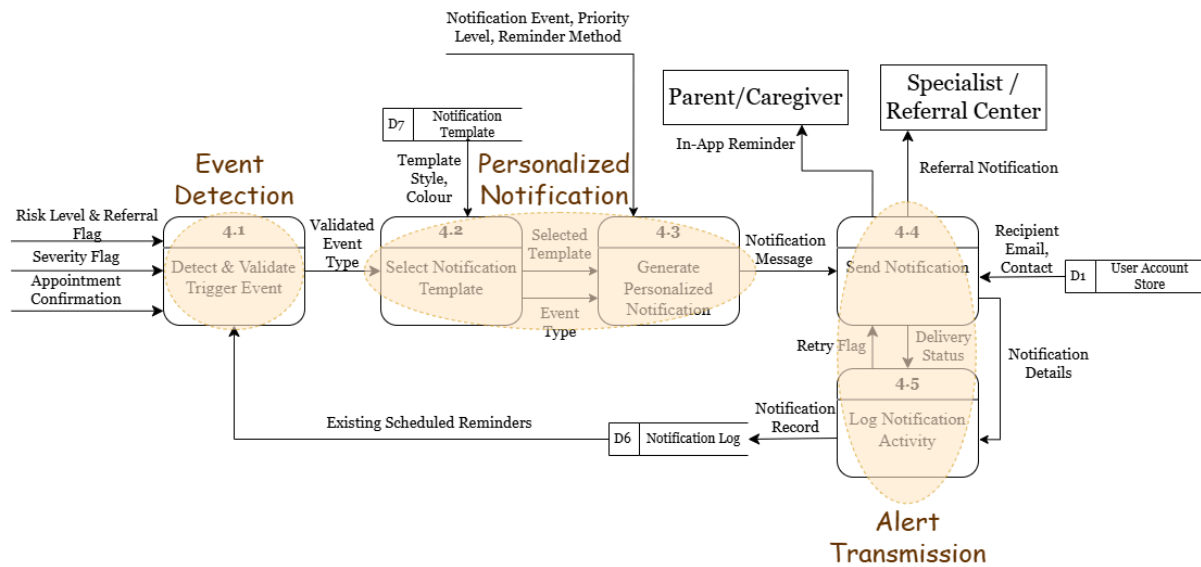
7.1.3.2.2 Child Diagram 2.0: BPPOKU Profile Management



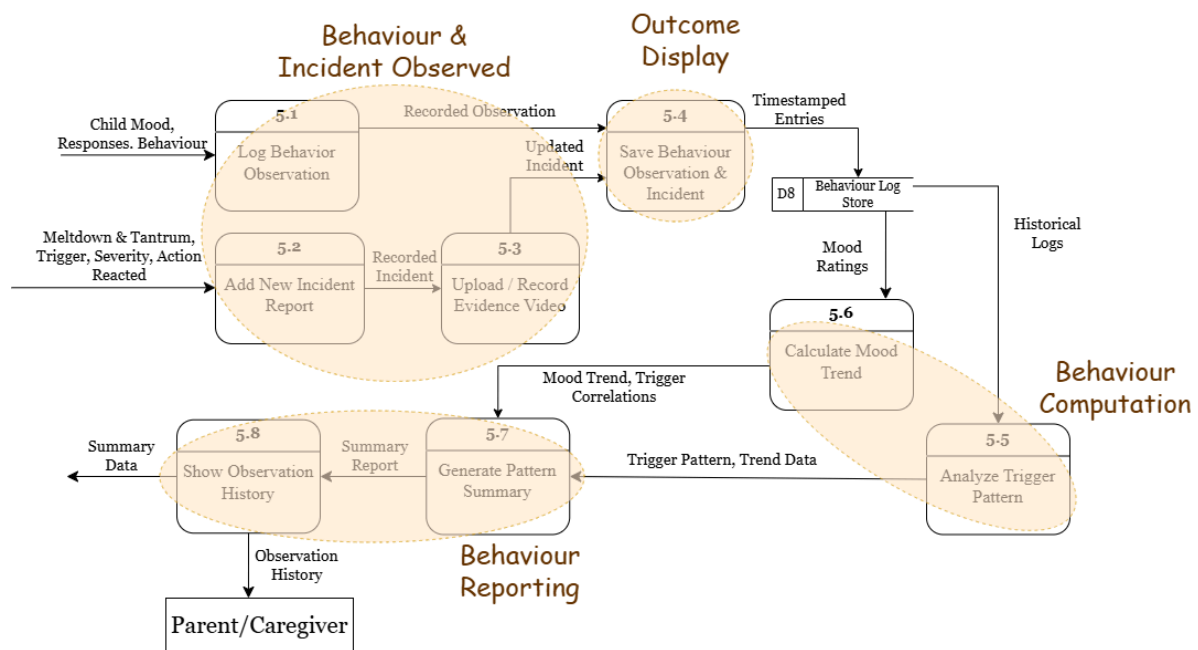
7.1.3.2.3 Child Diagram 3.0: M-CHAT-R Screening



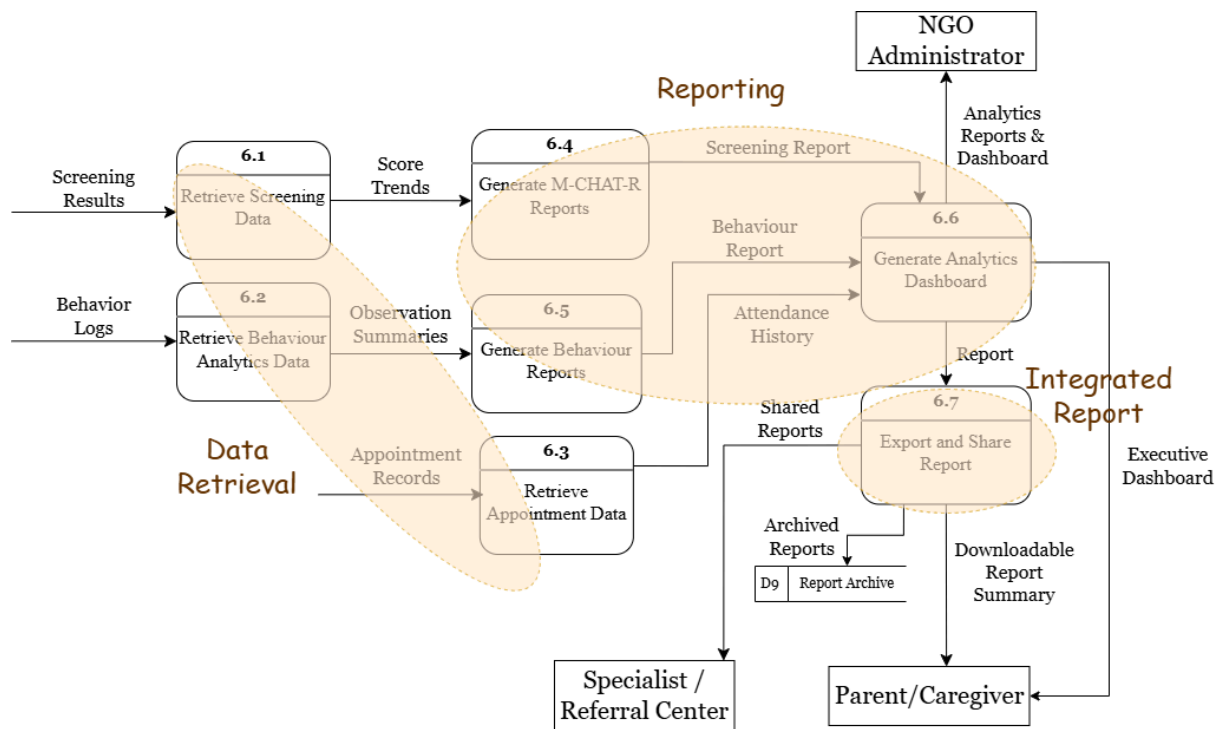
7.1.3.2.4 Child Diagram 4.0: Notification and Reminder Management



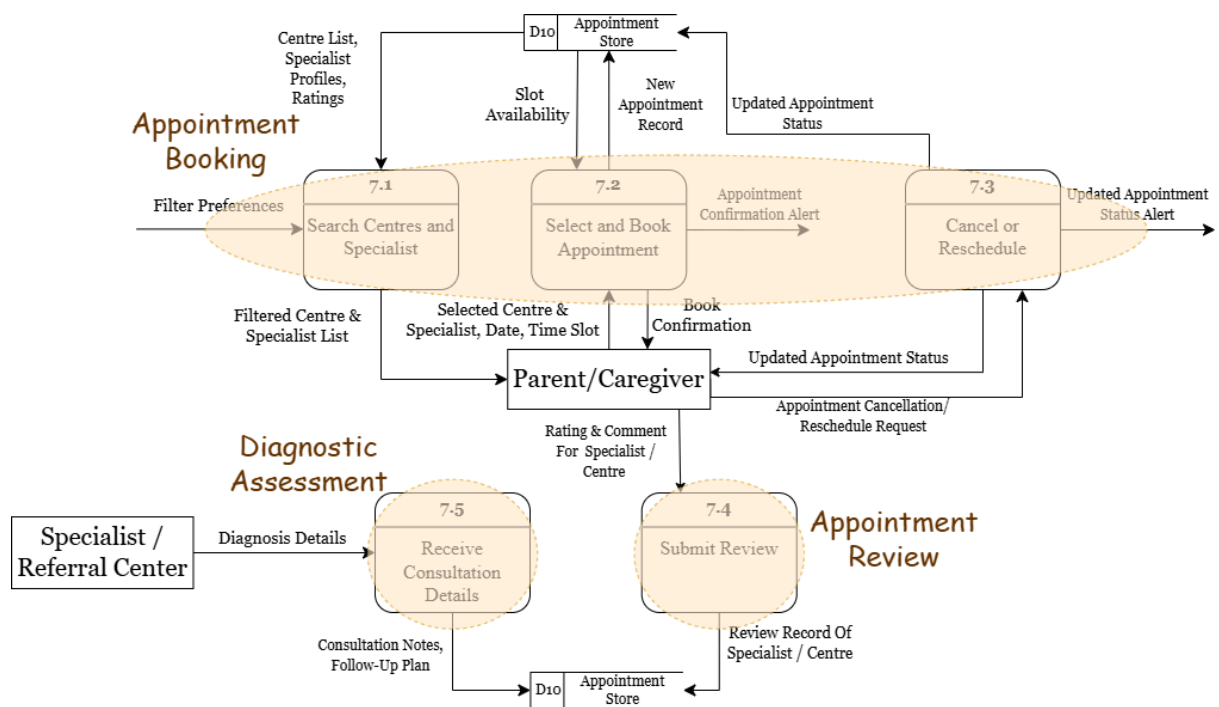
7.1.3.2.5 Child Diagram 5.0: Behaviour Monitoring



7.1.3.2.6 Child Diagram 6.0: Reporting and Analytics



7.1.3.2.7 Child Diagram 7.0: Appointment and Centre Management



7.2 CRUD Matrix

Activity	User Profile (D1)	Child Profile (D2)	Screening Result (D3)	Notification Log (D4)	Behaviour Log (D5)	Report (D6)	Appointment Store (D7)
Register User	C						
Login User	R						
Update User Profile	U						
Manage Child Profile		CRU					
Conduct M-CHAT-R Screening		R	CRU				
Manage Notification				CRUD			
Monitor Child Behaviour		R	R		CRUD	U	
Generate Analytics & Reports	R	R	R		R	CRU	R
Book Appointment		R					CRUD

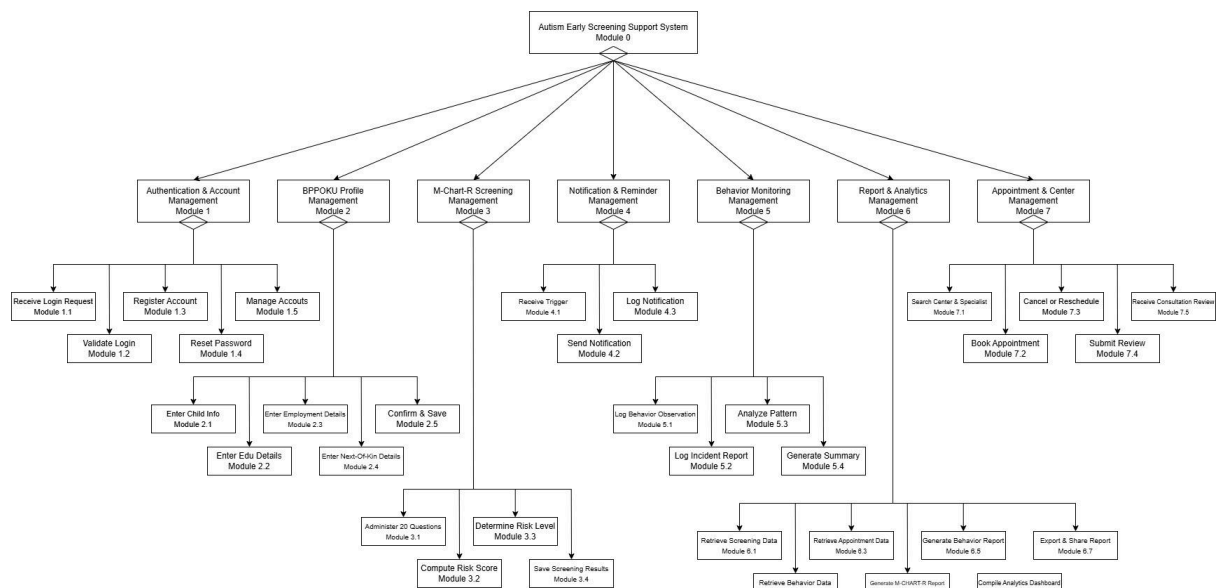
7.3 Event Response Table

Event	Source	Trigger	Activity	Response	Destination
User Registration	Parent / Caregiver	Parent clicks Sign Up	Create a new user account and validate registration details	User account created and registration confirmation displayed	Parent / Caregiver
User Login	Parent / Caregiver / NGO Administrator	User enters valid login credentials and clicks Login	Authenticate user credentials	User is redirected to the appropriate dashboard	Parent / Caregiver / NGO Administrator
BPPOKU Profile Submission	Parent / Caregiver	Parent submits BPPOKU profile form	Validate and save child profile information	Profile successfully saved	Parent / Caregiver
Start M-CHAT-R Screening	Parent / Caregiver	Parent selects Start Screening	Retrieve child information and display M-CHAT-R questionnaire	Screening questionnaire displayed	Parent / Caregiver
Submit M-CHAT-R Screening	Parent / Caregiver	Parent clicks Submit after completing the questionnaire	Calculate risk score and store screening result	Risk level and follow-up recommendation displayed	Parent / Caregiver
High-Risk Screening Detected	System	Screening result is classified as High Risk	Generate referral notification	Referral notification sent and recorded	Parent / Caregiver, Specialist / Referral Centre
Record Daily Behaviour Observation	Parent / Caregiver	Parent clicks Save Observation	Save behaviour observation into Behaviour Log	Observation successfully recorded	Parent / Caregiver
Submit Behaviour Incident	Parent / Caregiver	Parent clicks Save Incident	Store incident details and uploaded evidence	Incident report successfully saved	Parent / Caregiver

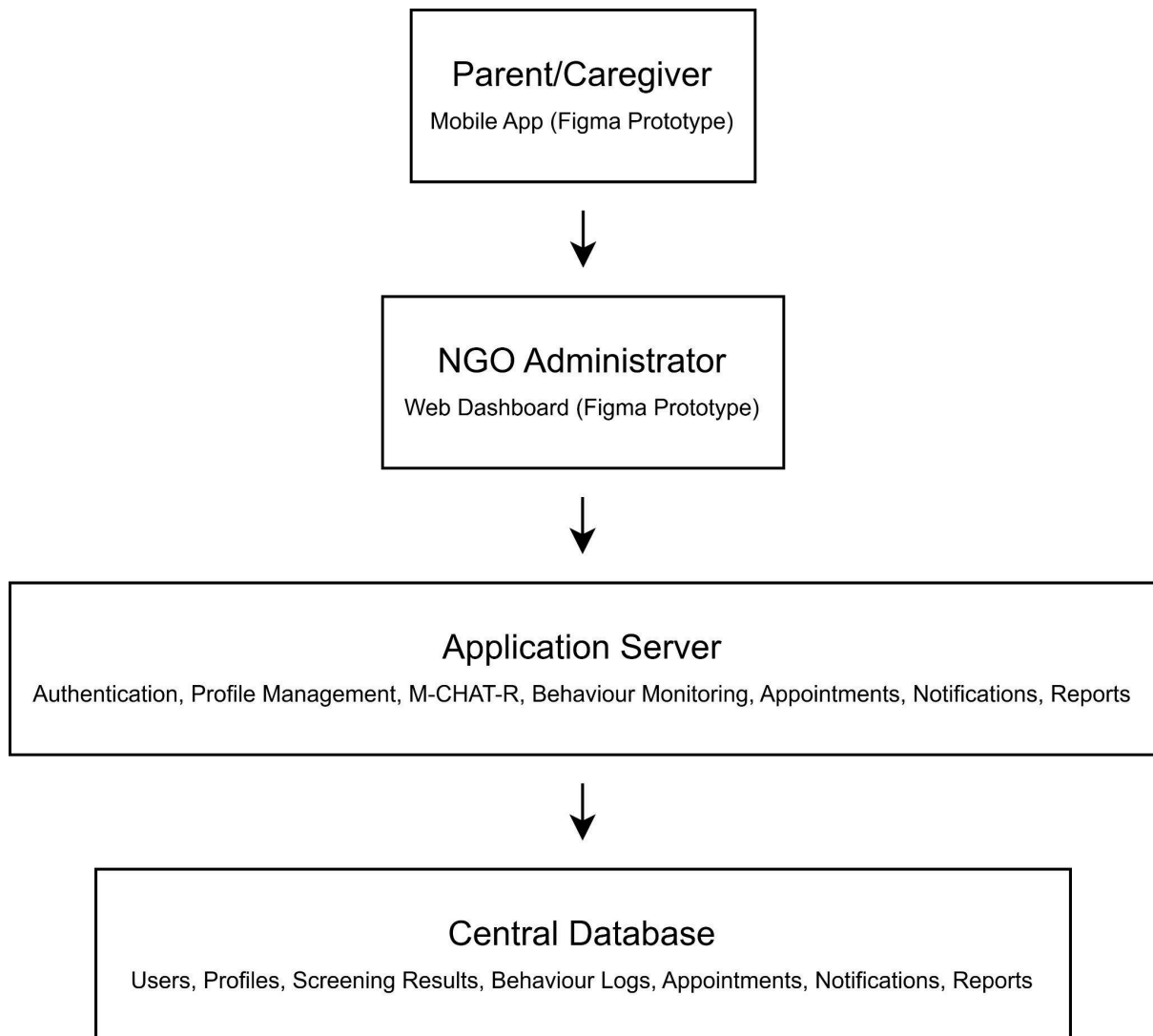
Book Appointment	Parent / Caregiver	Parent confirms appointment booking	Validate time slot and create appointment record	Appointment confirmation generated	Parent / Caregiver
Cancel or Reschedule Appointment	Parent / Caregiver	Parent submits cancellation or rescheduling request	Update appointment record	Updated appointment status displayed	Parent / Caregiver
Generate Reports	Parent / Caregiver / NGO Administrator	User selects Generate Report	Compile screening, behaviour, and appointment data	Report or analytics dashboard generated	Parent / Caregiver / NGO Administrator
Send Reminder Notification	System	Scheduled reminder time or system event occurs	Generate and dispatch notification	Reminder notification delivered and logged	Parent / Caregiver

Table 7.3.1 Event Response Table

7.4 Structure Chart



7.5 System Architecture



While designing the Autism Early Screening Support System, the three-tier architecture is applied. The first tier includes user interface components: mobile application for use by the caregivers or parents and the website for the NGO administration. The two tiers communicate with the application server via the Internet connection. The business logic components, including authentication, profile management, M-CHAT-R, behavior monitoring, appointments, notification, and reporting will take place in the application server. All system data is stored in a centralized database.

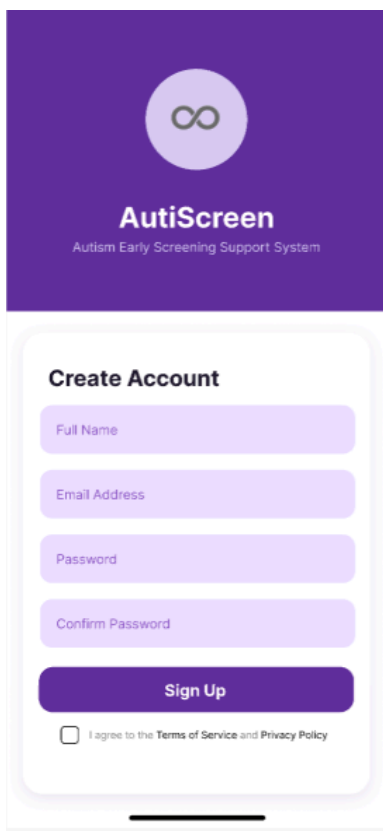
8.0 SYSTEM WIREFRAME

This part explains the architecture of UI wireframes that have been created for both parents and NGO who are responsible for the functioning of the Autism Early Screening Support System. The user interface has been designed according to the design requirements that were collected in the analysis phase, and it has been developed in a lively, clutter-free, and highly scannable way.

8.1 Parent Mode Dashboard Interface

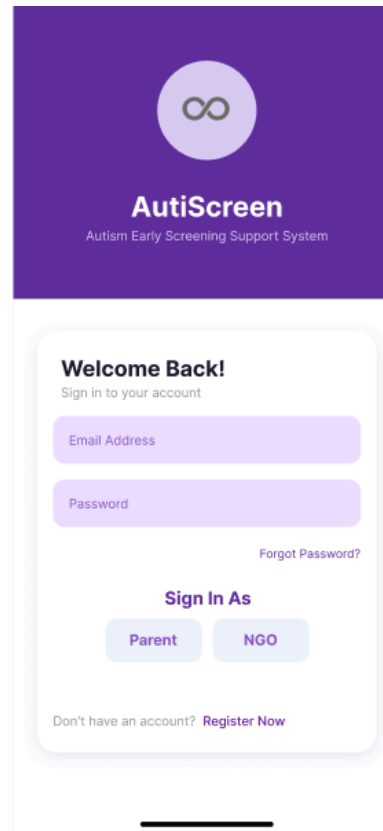
8.1.1 Sign Up and Sign In

Figure 8.1.1.1 shows the sign-up interface through which the users can gain secure access to the Autism Early Screening Support System. If users don't have an account, they can enter the registration information such as full name, email and password and then click on "Sign Up" once they agree with the terms and privacy policy of the platform. If users already have an account, the interface is transitioned into Sign In mode as shown in Figure 8.1.1.2. The user enters their email and password, selects their role-based account classification using the role selection buttons (Parent or NGO) and clicks on the login button for authentication purposes. Then, the system automatically redirects them to their corresponding role-based dashboard interface. Users who are not authenticated or need their credentials back can make use of the contextual action triggers (Forgot Password? or Register Now).



The wireframe for the Sign Up page features a purple header with the AutiScreen logo and title. Below the header, the 'Create Account' section contains four input fields: Full Name, Email Address, Password, and Confirm Password. A large purple 'Sign Up' button is positioned below these fields. At the bottom, there is a checkbox for agreeing to the Terms of Service and Privacy Policy.

Figure 8.1.1.1 Sign Up Page



The wireframe for the Sign In page features a purple header with the AutiScreen logo and title. Below the header, the 'Welcome Back!' section includes a sub-header 'Sign in to your account' and two input fields: Email Address and Password. A 'Forgot Password?' link is located below the Password field. The 'Sign In As' section contains two buttons: 'Parent' and 'NGO'. At the bottom, there is a link for 'Don't have an account? Register Now'.

Figure 8.1.1.2 Sign In Page

8.1.2 Dashboard Interface

Figure 8.1.2.1 shows the main parent dashboard interface, which is designed to function as a single navigation point to help the parent manage their child's early screening and intervention process. Once the user successfully logs into the application, they will get an automatically generated personal welcome message along with a profile card that includes their child's name, age, and current M-CHAT-R risk level. The quick access to the key tasks can be done using a 4-quadrant Quick Actions section with dedicated navigation blocks such as M-CHAT-R Screening, Reports & Analytics, Book Center Appointment, and Behaviour Monitor. Further, there are also two buttons under the category of Other Features for the Special Edu Centre Directory and the Community Group portal. There is a navigation menu at the bottom that allows the user to quickly move between Home, Screen, Profile, and Settings sub-menus.

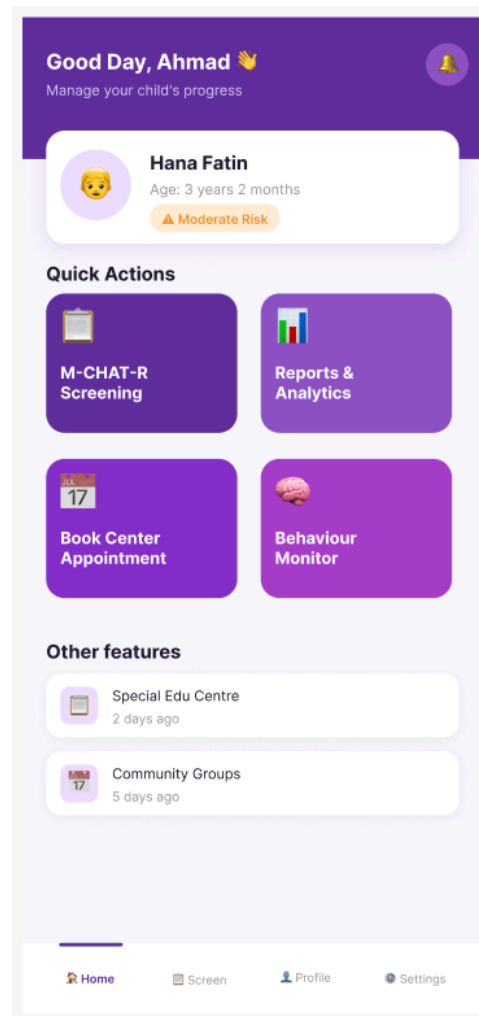


Figure 8.1.2.1 Parent Mode Dashboard Interface

8.1.3 Behavior Monitoring

When clicking the “Behavior Monitoring” button, Figure 8.1.12 will be shown. The system will display the current date. Parents can select between two buttons: “Daily Behavior Observation” routes to the diagnostic domain checklist, while “Meltdown & Tantrum Log” opens the acute incident reporting form.

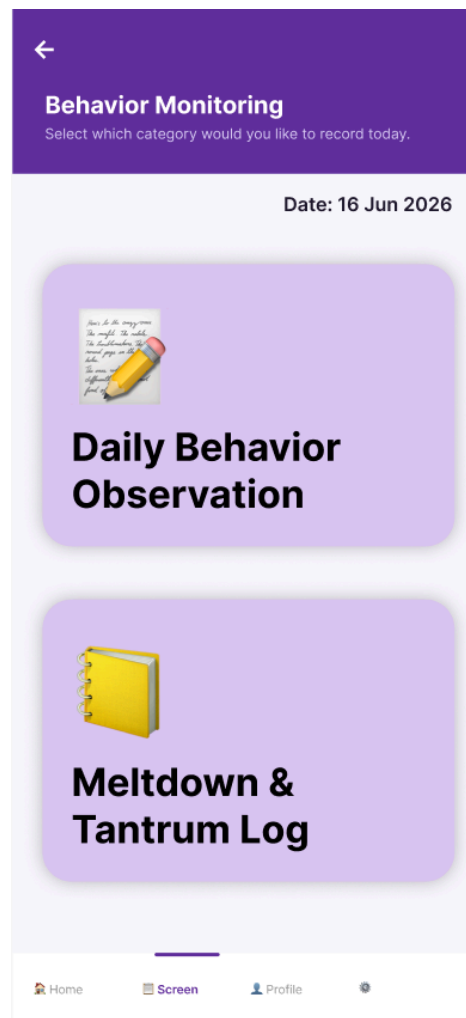


Figure 8.1.3.1 Behavior Monitoring Main Page

Figure 8.1.3.2 shows the screen after the parent has clicked the “Daily Behavior Observation” button. This form allows for selecting the baseline emotional state of the child depending on the type of day the child had. The form also provides a checklist for recording the clinical parameters for the child based on four main categories including Social Interaction, Communication, Sensory Sensitivities, and Repetitive and Restricted Behaviors. There is an additional text box where the parent can write down any extra information before clicking the “Save Observation” button or redirecting via the “View Observation History” utility button.

The figure displays two sequential screens of a mobile application titled "Daily Behavior Observation".

Left Screen:

- Title:** Daily Behavior Observation
- Subtitle:** Help us understand your child's behavior today.
- Date:** 16 Jun 2026
- Section:** How was your child's day?
- Options:** Good (😊), Neutral (😐), Difficult (😞)
- Instruction:** Select behaviors observed today.
- Categories and Checklists:**
 - Social Interaction:**
 - ☐ Avoids eye contact
 - ☐ Prefers playing alone
 - ☐ Does not respond to name
 - Communication:**
 - ☐ Repeats words or phrases
 - ☐ Difficulty understanding instructions
 - Sensory Sensitivities:** (Header visible, no checklist items shown)

Right Screen:

- Title:** Daily Behavior Observation
- Subtitle:** Help us understand your child's behavior today.
- Categories and Checklists:**
 - Communication:**
 - ☐ Repeats words or phrases
 - ☐ Difficulty understanding instructions
 - Sensory Sensitivities:**
 - ☐ Sensitive to loud sounds
 - ☐ Sensitive to touch
 - Repetitive & Restricted Behaviors:**
 - ☐ Hand flapping or repetitive movements
 - ☐ Strong interest in specific objects
 - ☐ Difficulty adapting to routine changes
- Additional Notes (Optional):**
 - Text input field: What happened before or after the behavior?
- Buttons:**
 - Save Observation
 - View Observation History

Figure 8.1.3.2 Daily Behavior Observation interface

Figure 8.1.3.3 shows the screen after clicking the “Save Observation” button while Figure 8.1.3.4 shows the Observation History Page. The display shows past data in the form of individual date cards, showing the particular characteristics logged as bulleted lists. Clicking on “View Pattern Summary” takes the parents to the analytical data visualization feature, whereas “Add New Observation” redirects them to the daily checklist pipeline.

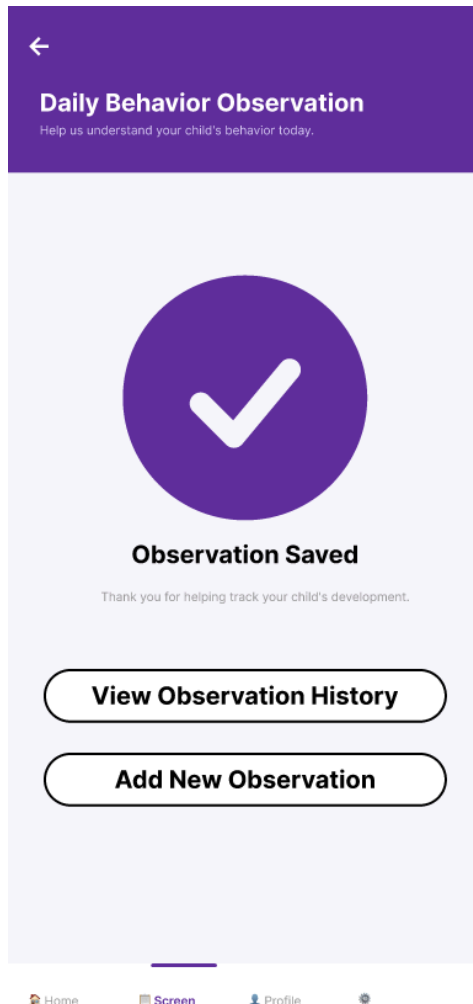


Figure 8.1.3.3 Observation Saved Page

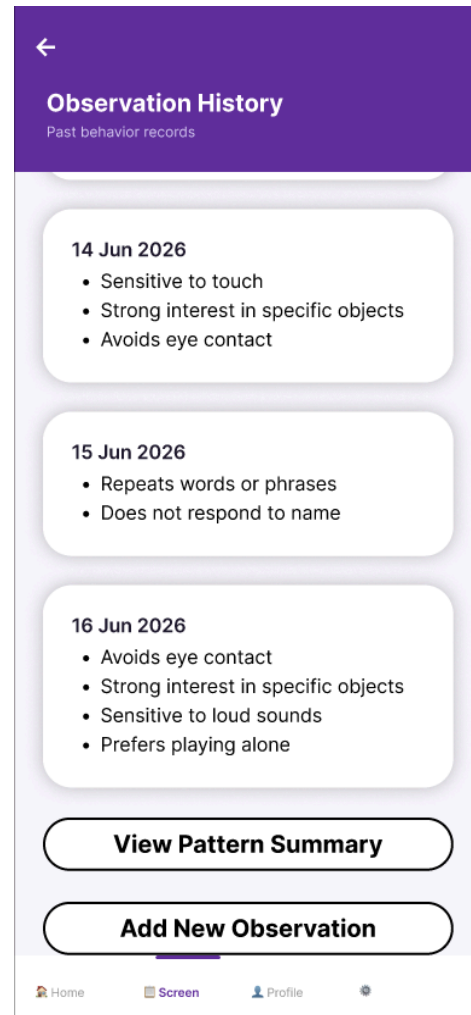


Figure 8.1.3.4 Observation History Page

Figure 8.1.3.5 shows the Behavioral Pattern Summary screen, which presents analytics based on tracking history for each week. The screen consolidates behaviors and generates metric cards that qualify as a Repeated Pattern, stating the exact frequency of the pattern along with explanatory text labels. After clicking the “View Report” button, the Behavior Report page is shown as Figure 8.1.3.6. It shows some important metrics. Parents can go further into more specific sections of analysis, which include Behaviour Observation Summary, Incident Triggers, Severity Analysis, and Most Recent Incidents Log. Primary action keys enable instant exportation of the document and creation of new incidents.

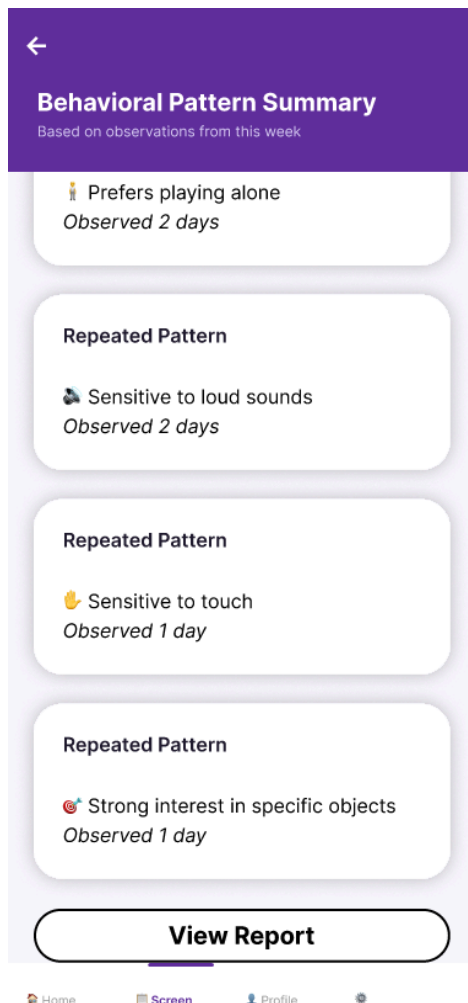


Figure 8.1.3.5 Behavior Pattern Summary

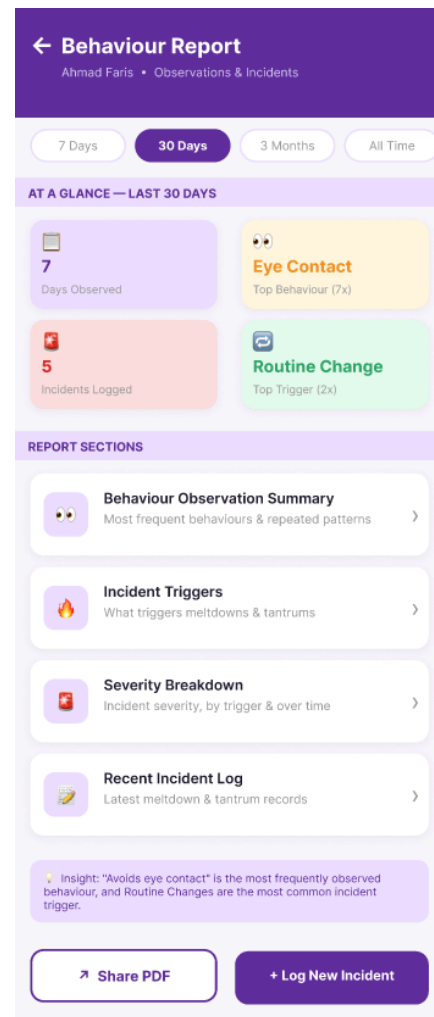


Figure 8.1.3.6 Behavior Report

Figure 8.1.3.7 shows the landing interface of the Meltdown & Tantrum Log, where previous behavioral emergencies are documented in the form of a chronology of color-coded history cards that list date, trigger, and clinical severity level. To log a current behavioral emergency, the parent taps on the "Add New Incident" action button that is anchored at the top of the interface as illustrated in Figure 8.1.3.8. The New Incident Report form opens up instantly and the runtime parameters such as Date, Time, and Duration are automatically populated by the system for the parent to fill in the environmental trigger, select the clinical severity level, tick off the observed behaviors, and enter situational text.

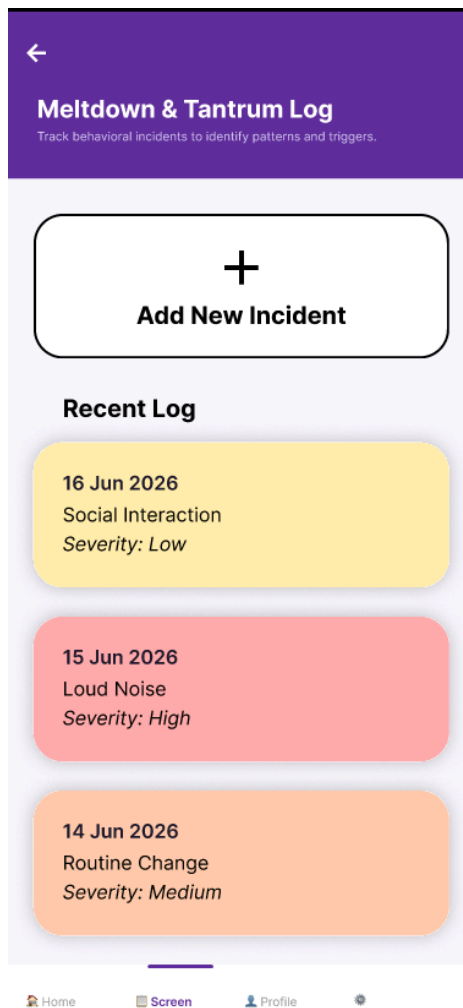


Figure 8.1.3.7 Meltdown & Tantrum Log
Page

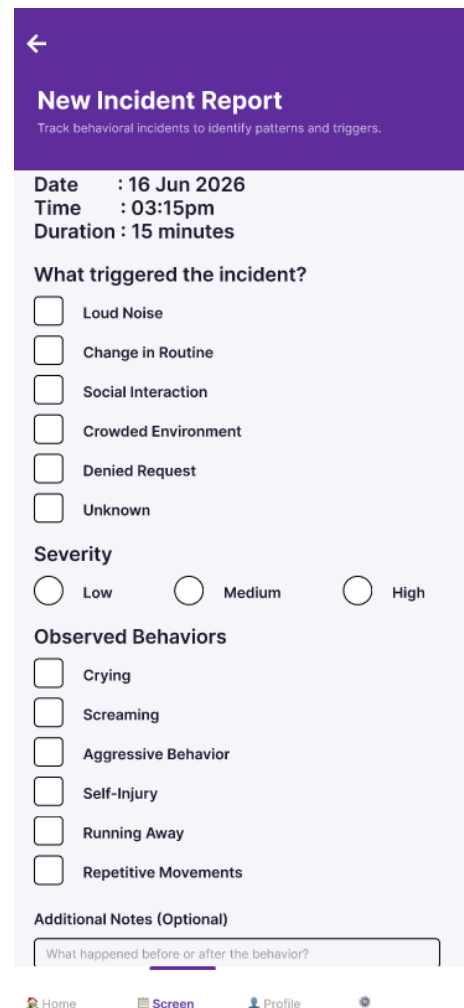


Figure 8.1.3.8 New Incident Report Page

Figure 8.1.3.9 shows the Behavior Evidence page that prompts parents to choose one of the two possible paths for uploading proof. Once the choice has been made, the system navigates to the validation screen as shown in Figure 8.1.3.10, showing the confirmation text together with the primary action buttons “Save Incident” or “Re-upload Video”.

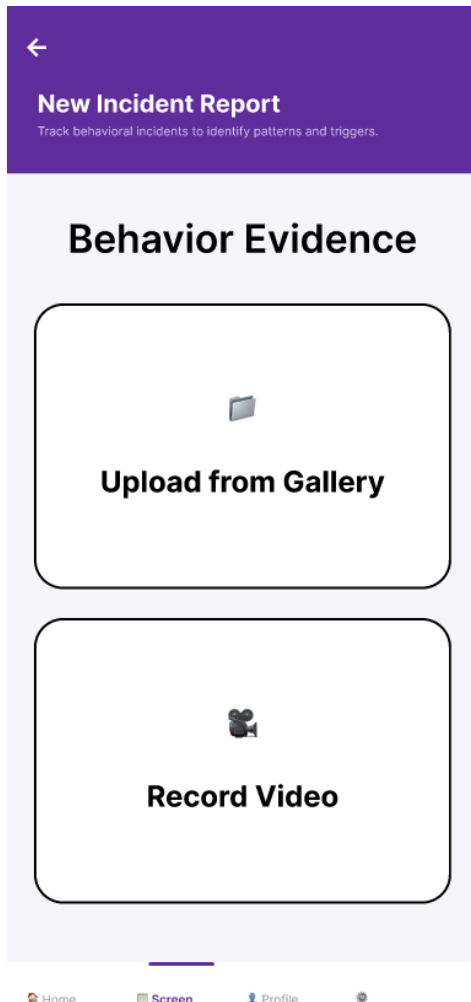


Figure 8.1.3.9 Behavior Evidence

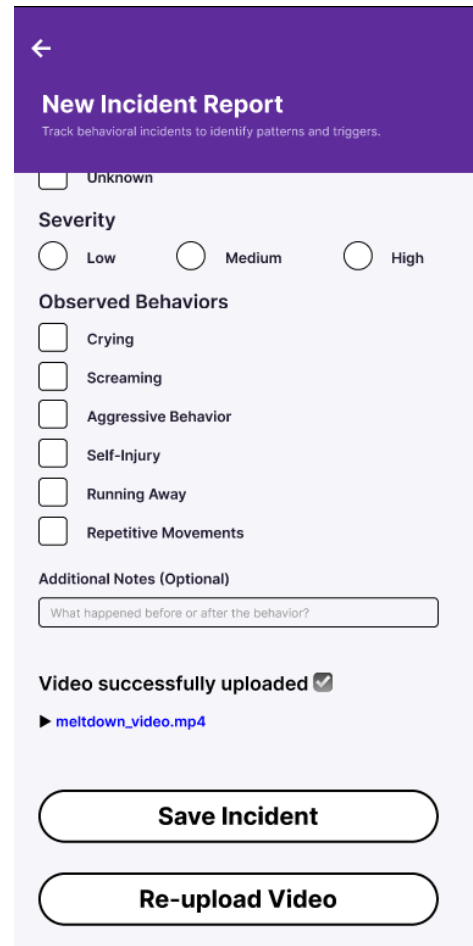


Figure 8.1.3.10 Video Uploaded Page

After the “Save Incident” button is clicked, the user is led to the success validation page as shown in Figure 8.1.3.11. Clicking the “View Incident History” button leads to the timeline dashboard as shown in Figure 8.1.3.12. It displays dates and details such as trigger causes, duration and severity of each crisis, along with the evidence.

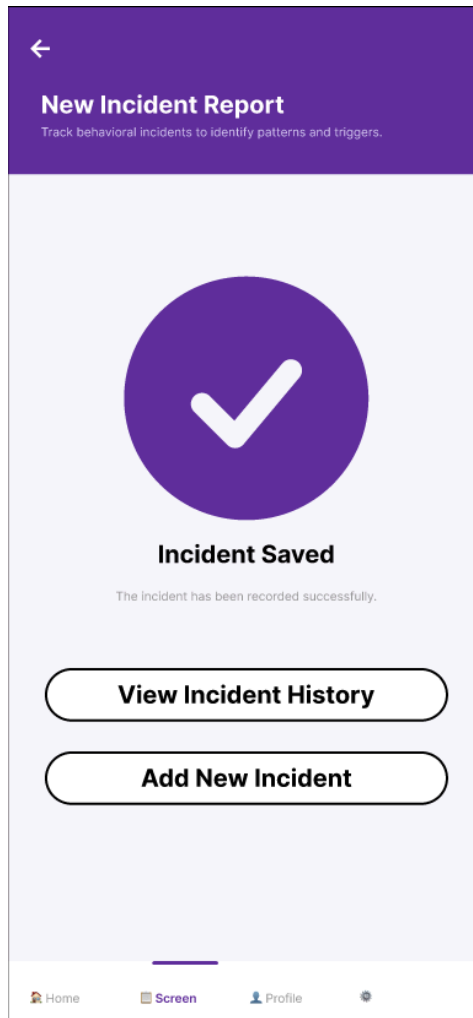


Figure 8.1.3.11 Incident Saved Page

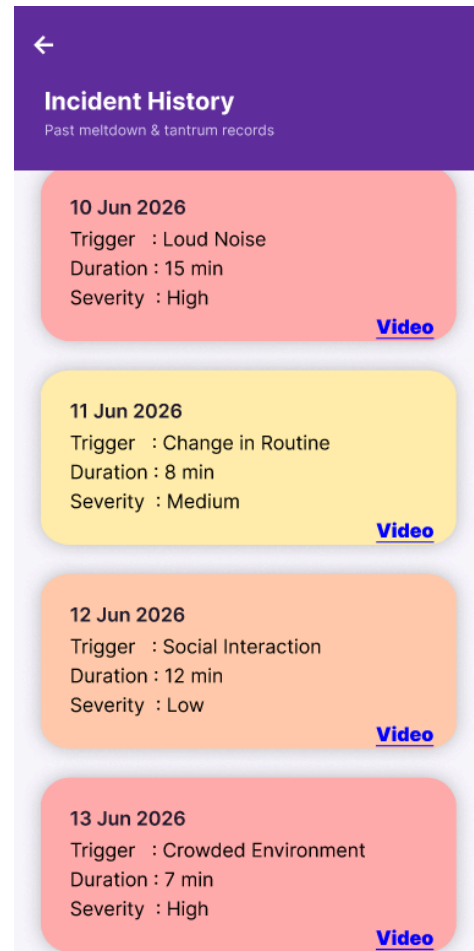


Figure 8.1.3.12 Incident History

8.1.4 Center Booking Appointment

The initial stage starts from the Find Centers directory as shown in Figure 8.1.4.1, where parents can search centers that are near to their residence, and click on "Book Center Visit" to redirect to the Center's details page as shown in Figure 8.1.4.2. In this page, parents get to know about the detailed information of a center. For the first-time user, it is suggested to click the "Let Center Match Specialist" button, which means the team will assign the specialist based on the child's needs when they arrive at the center. Parents can also click the message icon, which redirects them directly to the Consultation Chat Page as shown in Figure 8.1.4.3.

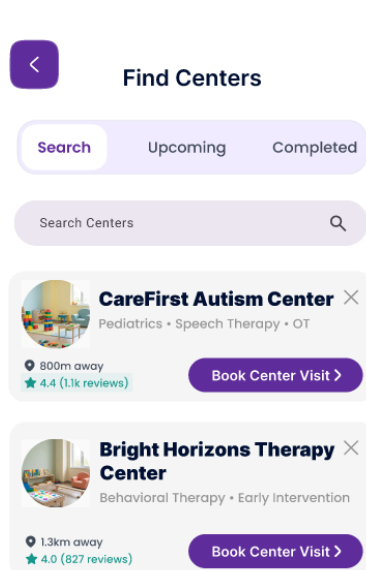


Figure 8.1.4.1 Find Centers Page

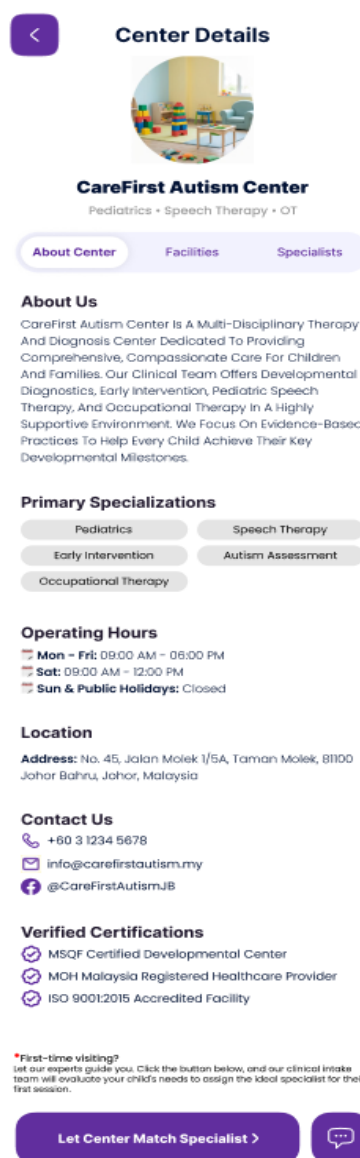


Figure 8.1.4.2 Center Details (About Center)

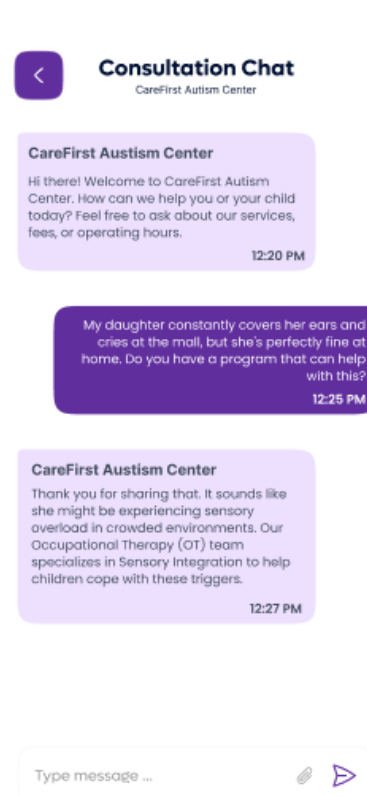


Figure 8.1.4.3 Consultation Chat

Upon clicking on the Facilities tab as shown in Figure 8.1.4.4, it will display the facilities available within the center's clinical framework. By switching to the Specialists tab as shown in Figure 8.1.4.5, it will display the specialists that work in the center

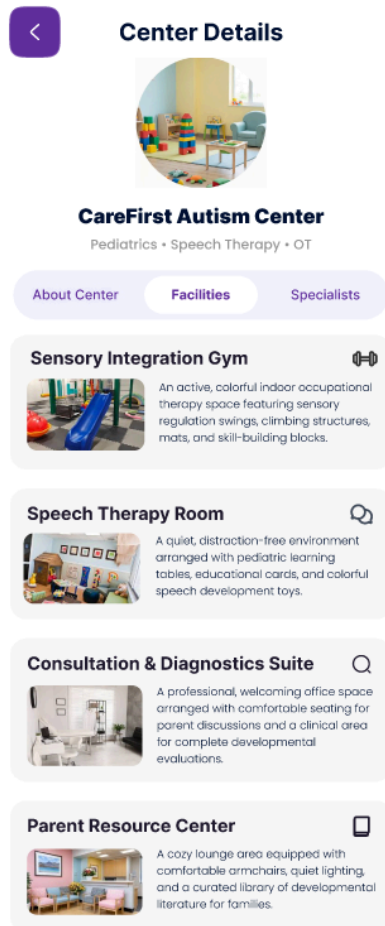


Figure 8.1.4.4 Center Details (Facilities)

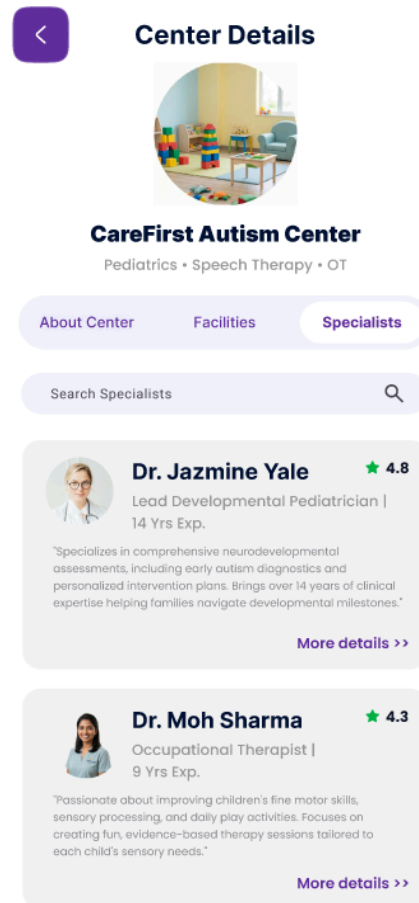


Figure 8.1.4.5 Center Details (Specialists)

Figure 8.1.4.6 shows the Specialist Profile, showing all of the background information for the respective specialist. Parents are required to select the date and time to confirm the appointment schedule as shown in Figure 8.1.4.7. After choosing the consultation type, it will display the appointment confirmation details page as shown in Figure 8.1.4.8. Parents can chat directly with the specialists, reschedule and delete the appointment by clicking on the button below.

Doctor Profile

Dr. Jazmine Yale
Lead Developmental Pediatrician

[Schedule](#) [About Doctor](#) [Review](#)

Select Date & Time

Sun 30 Mon 31 **Tue 1** Wed 2 Thu 3

08:00 09:00 **10:00** 11:00 12:00 13:00

Consultation Type

Video Call Walk-in

*Ensure a stable connection for your online consultation

[Book Appointment >](#) [Chat](#)

Figure 8.1.4.6 Specialist Profile

Appointment Schedule

Choose Date

< September 2026 > Month ^

S	M	T	W	T	F	S
30	31	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	01	02	03

Select Time Slot

09:00 AM	10:00 AM	11:00 AM
12:00 PM	01:00 PM	02:00 PM
03:00 PM	04:00 PM	05:00 PM

[Confirm >](#)

Figure 8.1.4.7 Appointment Schedule

Appointment Details

CareFirst Autism Center

Dr. Jazmine Yale
Developmental Pediatrician

Date & Time: 1 Sep 2026, 10:00 AM

Consultation Type: Video Consultation

Assigned Scheme: General Intake Assessment

Status: **Confirmed**

[Edit](#) [Delete](#) [Chat](#)

Figure 8.1.4.8 Appointment Confirmation Page

8.1.5 Help Center

Figure 8.1.5.1 shows the Help Center Interface. The screen is divided into three key sections: Contact Support which provides real-time assistance which allows users to reach out through chat, email, and phone line. Next is Quick Guides which provides step-by-step guides such as conducting M-CHAT-R screening, understanding results, and scheduling appointments. Last is Frequently Asked Questions where an accordion-type interface solves common issues about data protection and system use.

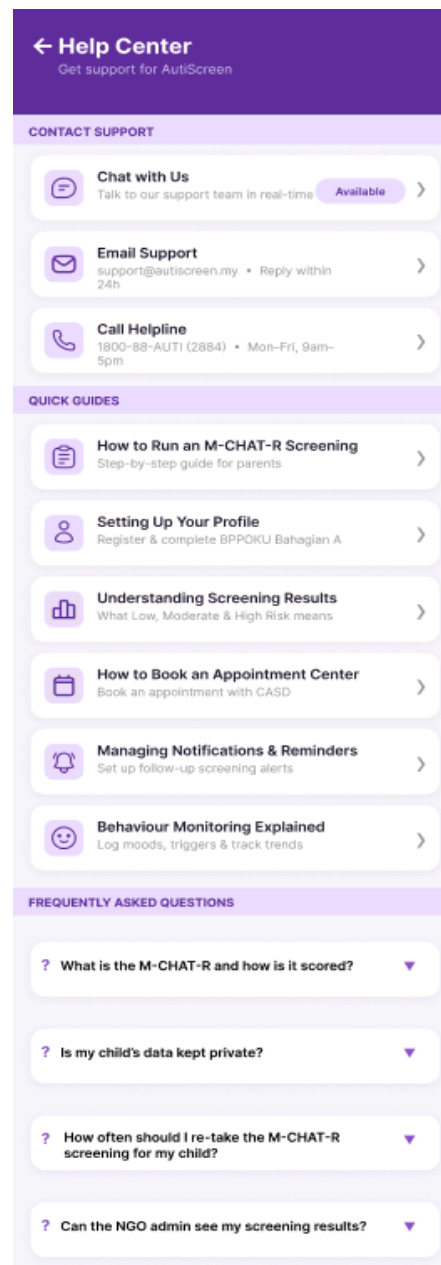


Figure 8.1.5.1 Help Center Interface

8.1.6 Special Edu Center

The Special Edu Center interface as shown in Figure 8.1.6.1, works as a local directory enabling parents to find nearby educational centers. It enables parents to search by area and choose between two types of centers: Private Centre which offers unique programs and flexible scheduling as shown in Figure 8.1.6.2 and Public Centre that offers government-funded and subsidized programs as shown in Figure 8.1.6.3.

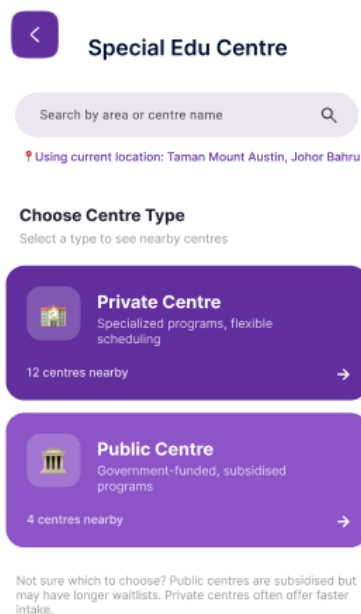


Figure 8.1.6.1 Special Edu Center Interface

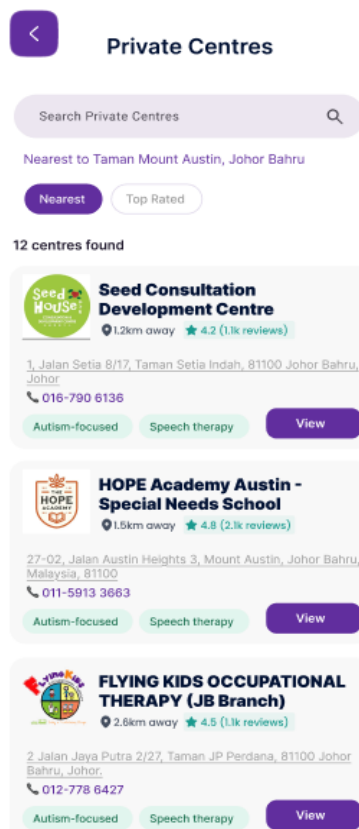


Figure 8.1.6.2 Private Centers

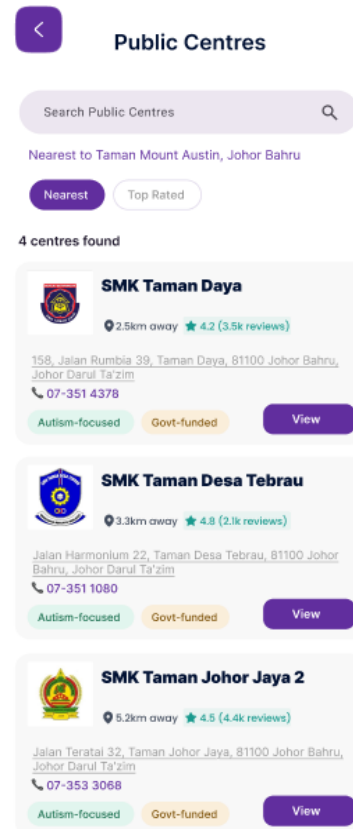


Figure 8.1.6.3 Public Centers

8.1.7 Community Groups

As can be seen in Figures 8.1.7.1 and 8.1.7.2, this is how the architecture of the peer-to-peer support network and community coordination works in this application. The process starts with the Community Groups feed from the global level as shown in Figure 8.1.7.1. It includes an index of parent communities filtered according to the targeted interest. In each community card, one can see how many subscribers there are currently and what is the level of activity in the form of live counters. Once the Join Group control button is pressed, users will enter the Discussion Stream as shown in Figure 8.1.7.2.

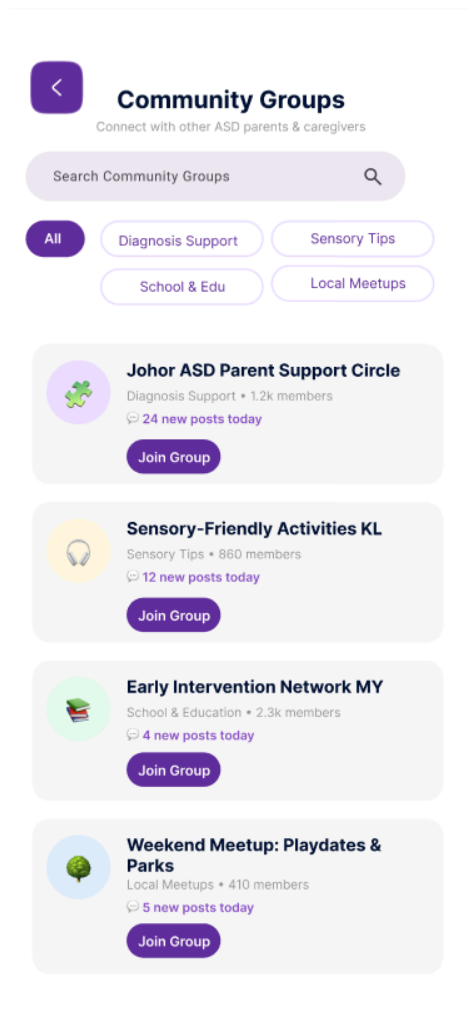


Figure 8.1.7.1 Community Groups

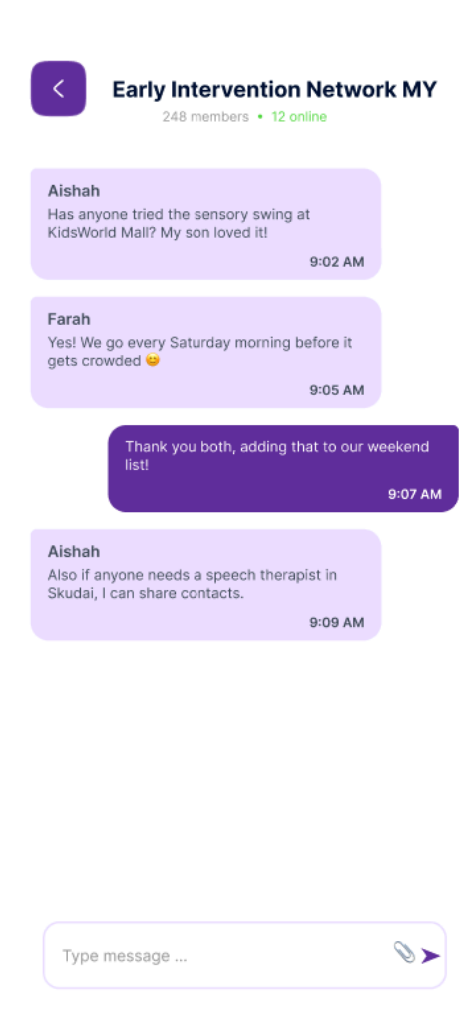


Figure 8.1.7.2 Discussion Stream

8.1.8 Profile Management

Figure 8.1.8.1 shows the main Account Profile interface. Parents are able to log out by clicking the “Sign Out” button at the bottom of the main profile page as shown in Figure 8.1.8.2.

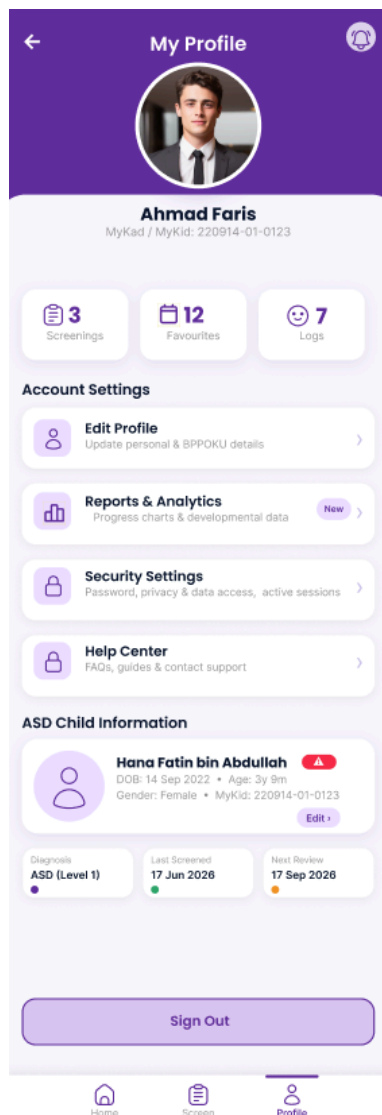


Figure 8.1.8.1 Main Account Profile

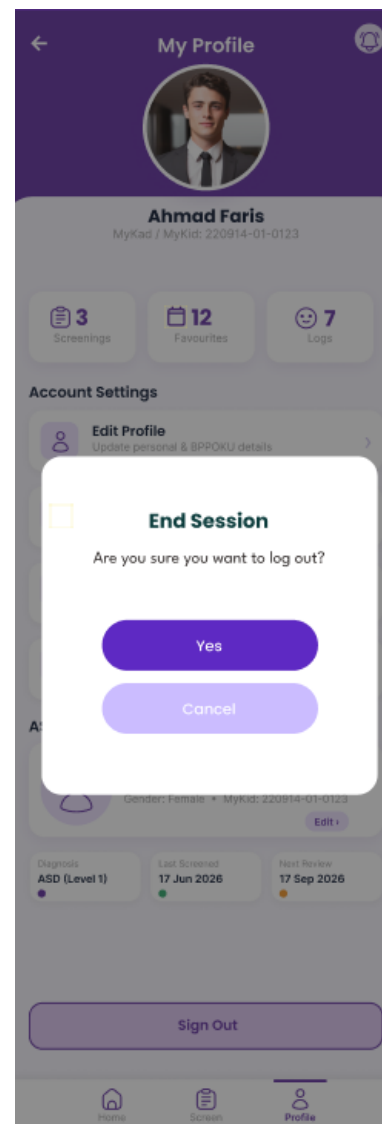


Figure 8.1.8.2 Sign Out Page

Figure 8.1.8.3 and Figure 8.1.8.4 shows the personal details and ASD child information details based on the BPPOKU Pindaan 2019. Parents are required to fill in all the demographic details.

← Edit Profile

1 Personal 2 Education 3 Employment 4 Next of Kin

PERSONAL INFORMATION

MyKad / MyKad No.
e.g. 010914-01-0123

Military IC No. (if applicable)
-

Full Name (as per MyKad / MyKID)
e.g. Ahmad Fatin bin Abdullah

Marital Status
☐ Single ☐ Married ☐ Widowed ☐ Divorced ☐ Separated

Number of Children
e.g. 2

Home Phone No.
e.g. 07-123 4567

Office Phone No.
e.g. 07-987 6543

Mobile Phone No.
e.g. 012-345 6789

Email Address
e.g. name@gmail.com

Mailing Address
No., Street, Area...
e.g. Skudai

Town / City
e.g. Skudai

Postcode
e.g. 81300

State
e.g. Johor

REGISTRATION INFORMATION

Registration Type
☐ New Registration ☐ Re-Registration
☐ Category Change ☐ Card Replacement

Previous Registration No. (if any)
-

Registration Date
DD / MM / YYYY

Ensure the MyKad/MyKID number matches the applicant's official identity document before proceeding.

Save & Next →

← Edit Profile

1 Personal 2 Education 3 Employment 4 Next of Kin

EDUCATION LEVEL

Highest Education Level (select one)
☐ No Schooling ☐ Pre-School ☐ Primary School ☐ Secondary School ☐ Vocational / Technical College ☐ Certificate ☐ Diploma ☐ Bachelor's Degree ☐ Master's Degree ☐ PhD / Doctorate

Field of Study (if applicable)
e.g. Computer Science

Name of Institution
e.g. Universiti Teknologi Malaysia

Year of Graduation / Completion
e.g. 2023

← Back **Save & Next →**

← Edit Profile

1 Personal 2 Education 3 Employment 4 Next of Kin

EMPLOYMENT SECTOR

Employment Sector (select one)
☐ Government ☐ Statutory Body ☐ Private ☐ Non-Governmental Organisation (NGO) ☐ Self-Employed ☐ Non-Applicable

EMPLOYMENT STATUS

Current Employment Status (select one)
☐ Unemployed ☐ Employer ☐ Government Employee ☐ Private Sector Employee ☐ Self-Employed ☐ Retiree / Pensioner

JOB DETAILS

Occupation / Job Title
e.g. Software Engineer

Employer / Company Name
e.g. ABC Technology Sdn. Bhd.

Employee Phone No.
e.g. 03-1234 5678

Employer Address
No., Street, Area...
e.g. Johor Bahru

Town / City
e.g. Johor Bahru

Postcode
e.g. 80000

State
e.g. Johor

MONTHLY INCOME

Monthly Household Income (RM)
☐ Below RM500 ☐ RM500 - RM999 ☐ RM1,000 - RM1,999 ☐ RM2,000 - RM2,999 ☐ RM3,000 - RM3,999 ☐ RM4,000 and above

Other Income Sources (e.g. Pension, PERKESO)
-

ADDITIONAL INFORMATION

Are you currently seeking employment?
☐ Yes ☐ No

Household income information is used to determine eligibility for assistance programmes offered by the Welfare Department.

← Back **Save & Next →**

← Edit Profile

1 Personal 2 Education 3 Employment 4 Next of Kin

NEXT OF KIN IDENTITY

MyKad No. (Next of Kin)
e.g. 850614-01-5678

Passport No. (if non-Malaysian)
-

Full Name (as per MyKad / MyKID)
e.g. Emily Zara binti Abdullah

Nationality
☐ Malaysian ☐ Non-Malaysian (specify below)
Specify Nationality (if Non-Malaysian)
-

RELATIONSHIP TO APPLICANT

Relationship (select one)
☐ Mother ☐ Father ☐ Spouse ☐ Guardian ☐ Sibling ☐ Other (specify)
Specify Relationship (if Other)
-

EMPLOYMENT & INCOME

Occupation
e.g. Teacher

Monthly Income (RM)
☐ Below RM500 ☐ RM500 - RM999 ☐ RM1,000 - RM1,999 ☐ RM2,000 - RM2,999 ☐ RM3,000 - RM3,999 ☐ RM4,000 and above

PERMANENT ADDRESS

Address
No., Street, Area...
e.g. Skudai

Town / City
e.g. Skudai

Postcode
e.g. 81300

State
e.g. Johor

Mailing Address (if different from above)
No., Street, Area...
e.g. Skudai

Town / City
e.g. Skudai

Postcode
e.g. 81300

State
e.g. Johor

CONTACT INFORMATION

Home Phone No.
e.g. 07-123 4567

Office Phone No.
e.g. 07-987 6543

Mobile Phone No.
e.g. 012-345 6789

Email Address
e.g. name@gmail.com

Please verify all information
Errors or omissions are accurate and up-to-date before submitting the registration form.

← Back **Submit Profile**

Figure 8.1.8.3 Personal Details Profile

← ASD Child Information
Personal details of your registered child

1 Personal 2 Diagnosis 3 Support & Intervention

CHILD IDENTITY

Full Name (as per MyKID)
e.g. Hana Fatin binti Abdullah

MyKID No.
e.g. 220914-01-0123

Date of Birth
DD / MM / YYYY

Age (auto-calculated)
3 years 9 months

Gender
☐ Male ☐ Female

Citizenship
☐ Malaysian ☐ Non-Malaysian

Race / Ethnicity
e.g. Malay

Religion
e.g. Islam

Birth Certificate No.
e.g. JHR-2022-0001234

RESIDENTIAL ADDRESS

Address Line 1
No., Street Name...

Address Line 2 (if applicable)
Area / Taman...

Town / City
e.g. Skudai

Postcode
e.g. 81300

State
e.g. Johor

Save & Next →

← ASD Child Information
Personal details of your registered child

1 Personal 2 Diagnosis 3 Support & Intervention

ASD DIAGNOSIS DETAILS

Primary Diagnosis
e.g. Autism Spectrum Disorder (ASD)

ASD Level / Severity (DSM-5)
☐ Level 1 - Requiring Support
☐ Level 2 - Requiring Substantial Support
☐ Level 3 - Requiring Very Substantial Support

Diagnosing Specialist / Clinic
e.g. Dr. Amir - Hospital Sultanah Aminah

Date of Diagnosis
DD / MM / YYYY

OKU Registration No. (if any)
e.g. OKU-2023-001234

Co-occurring Conditions (select all that apply)
☐ Speech Delay ☐ Intellectual Disability ☐ ADHD ☐ Anxiety Disorder ☐ Epilepsy / Seizures ☐ Sensory Processing Disorder

Other Conditions (specify)
-

← Back **Save & Next →**

← ASD Child Information
Personal details of your registered child

1 Personal 2 Diagnosis 3 Support & Intervention

CURRENT SUPPORT & INTERVENTION

Current Therapies Received (select all that apply)
☐ Speech Therapy ☐ Applied Behaviour Analysis (ABA) ☐ Special Education (Pendidikan Khas) ☐ Occupational Therapy ☐ Early Intervention Programmes (EIP) ☐ Other

Name of Therapy Centre / School
e.g. Pusat Pemulihan Dalam Komuniti Skudai

Therapist / Teacher Name
e.g. Cik Nurul Ain

Session Frequency
e.g. 3x per week

Diagnosis details are used solely to personalise screening and support recommendations. All data is encrypted and kept confidential.

Back **Save Child Info**

Figure 8.1.8.4 ASD Child Information Details

8.1.9 Screening (M-CHAT-R Process)

An illustrative representation of the interactive 20-question M-CHAT-R screening questionnaire interface design employed during the screening process lifecycle can be found in Figure 8.1.9.1. Parents are required to answer all 20 developmental milestone questions by clicking the “YES” or “NO” button. In order to maintain operational transparency, a historical summary node displays past committed question chips directly on top of the risk index tracker in real-time.

The screenshot displays the M-CHAT-R Screening interface. At the top, a purple header bar contains a back arrow, the title "M-CHAT-R Screening", the subtitle "Checklist for Autism in Toddlers", and a progress indicator "Question 20 of 20". The main content area is white with a purple border. It features a question card for question 20: "Does your child like movement activities?" with a subtext "For example, being swung or rocked on your lap?". Below the question are two buttons: a purple "✓ YES" button and a white "× NO" button. Underneath the buttons is a "Previous Answers" section showing a grid of 20 colored circles representing the status of each question. A legend below the grid defines the risk levels: Low (0-2) in green, Moderate (3-7) in orange, and High (8-20) in red. The current score is 16, and the status is High Risk. A "Submit" button is located at the bottom right of the main content area. At the very bottom, a navigation bar shows icons for Home, Screen, Profile, and a settings gear.

←
M-CHAT-R Screening
Checklist for Autism in Toddlers
Question 20 of 20

20
Does your child like movement activities?
For example, being swung or rocked on your lap?

✓ YES × NO

Previous Answers

1 2 3 4 5 6 7 8 9 10
11 12 13 14 15 16 17 18 19 20

Risk Levels:
● Low (0-2) ● Moderate (3-7) ● High (8-20)
Current Score: 16 | Status: High Risk

Submit

Home Screen Profile

Figure 8.1.9.1 One of the 20 Questions of the M-CHAT-R Screening Questionnaire

On completion of the Submit button press, the validation system determines the number of accumulated warning signs out, and displays the automated screening risk summary page based on the scores as shown in Figure 8.1.9.2. It will also display several recommended steps based on the result scores. Parents get to see their categorization analysis and have the option to view the report or share it with the healthcare professionals.

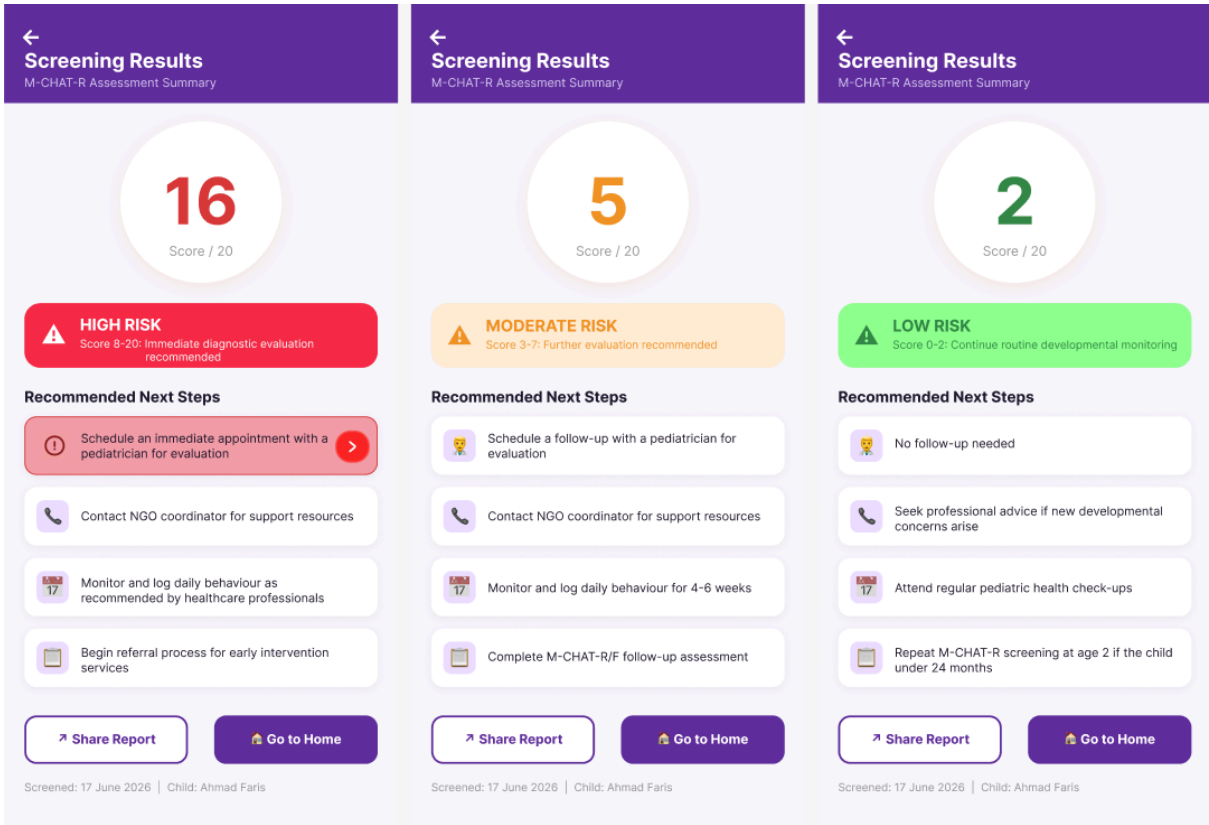


Figure 8.1.9.2 Screening Results (High, Moderate and Low Risk)

8.1.10 Notification

This process begins with OS-level push notifications presented on the device lock screen as shown in Figure 8.1.10.1, which contain highly important timely notifications such as a warning about high-risk screening results or behavioral logging prompts. When clicking on any message, the user gets redirected to the in-app Notification Center as shown in Figure 8.1.10.2, which contains notifications sorted chronologically into clearly identified categories. In order to limit the number of alerts and prevent users from being notified too often, users can enter the Reminder Frequency module as shown in Figure 8.1.10.3 in order to customize reminder intervals or choose a custom day and time frame.

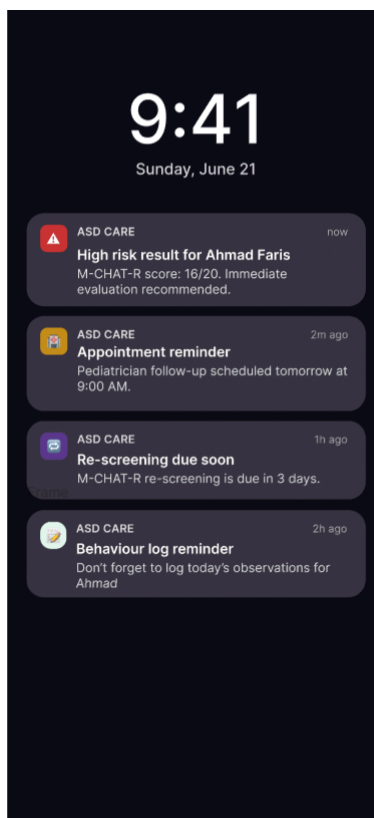


Figure 8.1.10.1 Lock Screen Notification

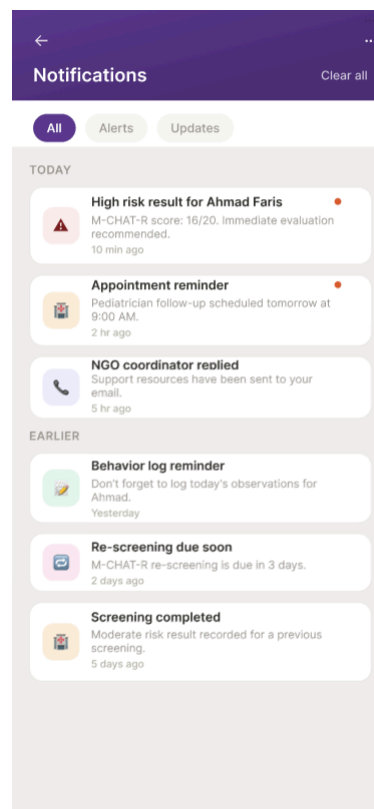


Figure 8.1.10.2 In-app Notification Center

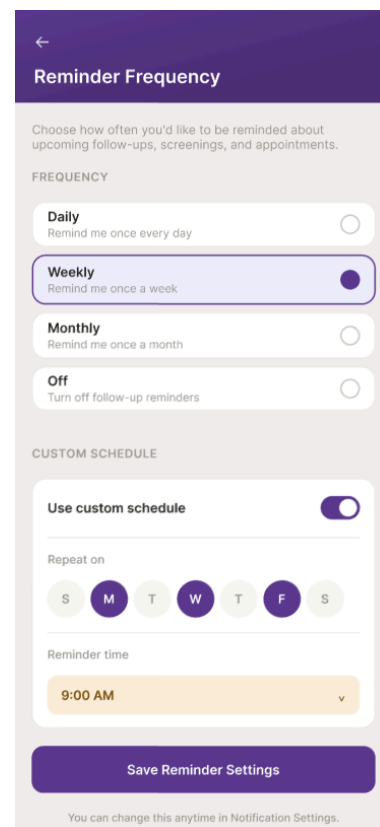


Figure 8.1.10.3 Reminder Frequency

8.1.11 Reporting and Analytics

Figure 8.1.11.1 shows the Reporting and Analytics dashboard. It is designed as a centralized historical database, which collects data automatically by integrating quantitative telemetry from M-CHAT-R diagnostics screening processes and behavioral logging to create a comprehensive progress profile of the chosen child. This visualization approach includes a hierarchically organized structure that consists of data matrices, individual reports, and data export functions. This approach traces longitudinal trends to inform parents about current risks and behaviors before sharing the information with health professionals.

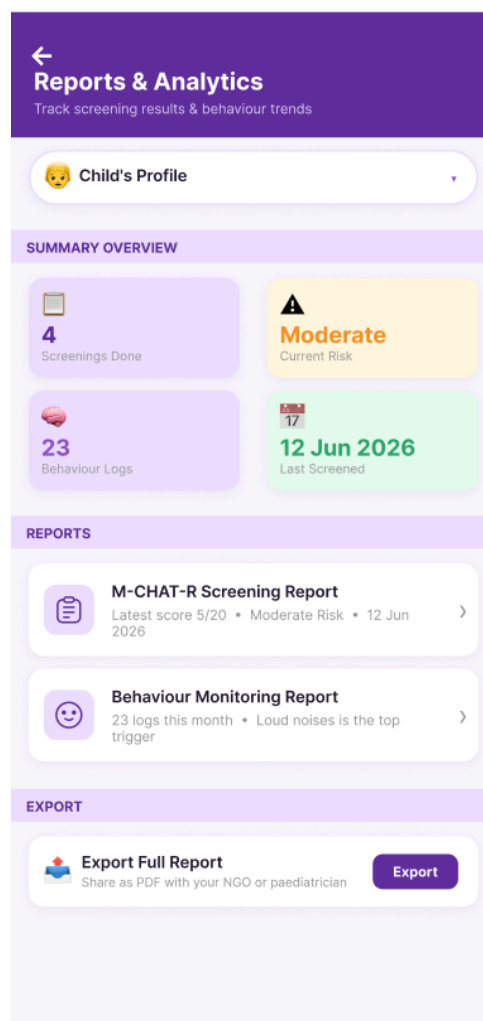


Figure 8.1.11.1 Reports & Analytics Page

8.2 NGO / Admin Mode Dashboard Interface

The centralized administration command web portal (AutoScreen Portal) as shown in Figure 8.2.1 serves as a secure multi-tenant data repository for non-governmental organizations and regional social welfare administrators. The tool automatically collates screening data quantitatively from single mobile applications for operational visibility of case registries. Such a centralized operation facilitates regional management through development of case profiles, prioritization of cases that require immediate attention, and implementation of intervention strategies in local centers.

The top core metrics row offers a volumetric analysis of the caseloads of the network. This row includes a Total Screenings Pending Card for unverified screenings, a Total Registered Children Card for number of platform directory registrants, a Flagged / High Risk Cases Card for clinical alert cases requiring immediate medical action, and a Total Sponsored Children Card for active intervention subsidy distributions within underserved families.

There are also grids which organize data on live operational and scheduling processes. The Screening Request Queue Table presents a live stream of the parent requests arranged according to the applicant's age and scores in addition to priority badges colored according to high risk, moderate risk, and low risk in addition to inspection anchors to facilitate auditing process. This queue works alongside the two variables Patients Data Trends Chart which monitors the monthly registration numbers in addition to the Latest Appointments Table.

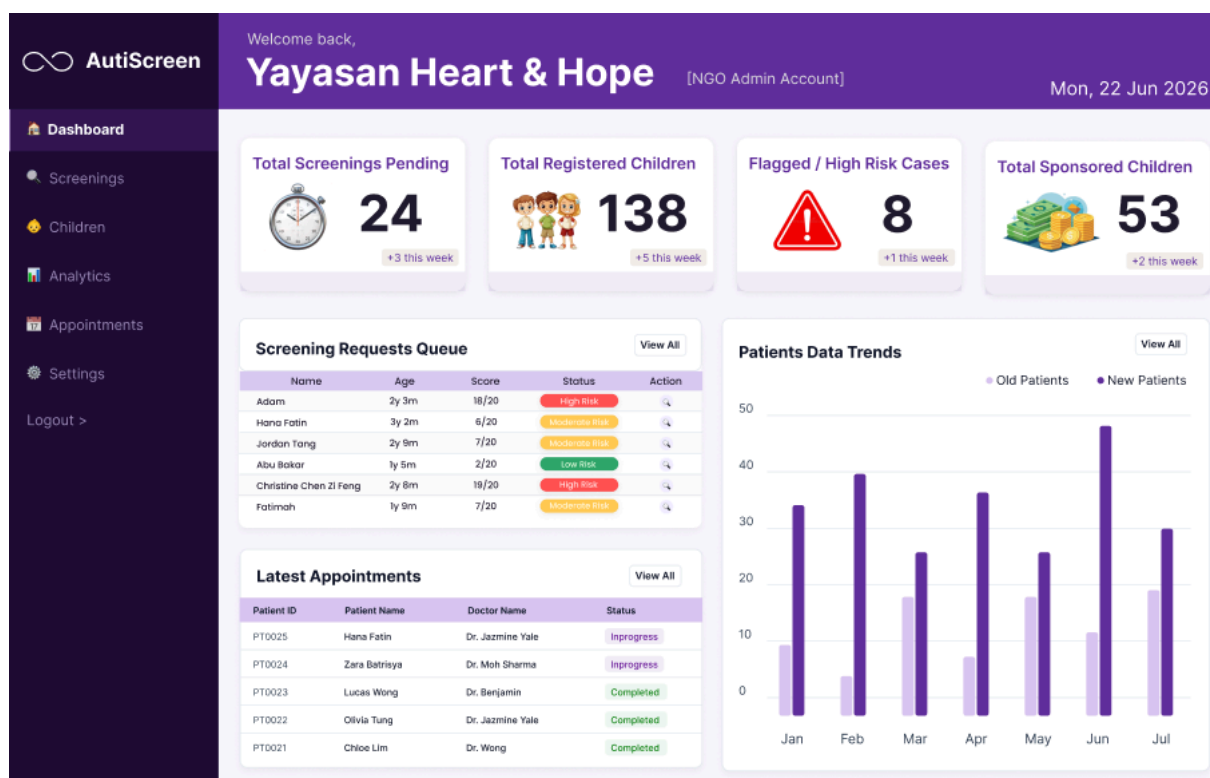


Figure 8.2.1 Dashboard

Secondary analysis of the administrative portal consists of the high-density grid of data visualization which encompasses clinical telematics and socio-economic metrics of the registry. This grid converts the localized screening data to structural demographic data that enable policy coordinators and non-governmental organizations to assess the trend of risk within communities and organize the distribution of healthcare and education resources systematically.

The top grid analyzes diagnostic and gender distribution among the pediatric registry. The Breakdown by Diagnostic Need horizontal chart identifies the primary care demand distributed across four core diagnoses: Autism Spectrum Disorder (ASD), Speech & Language Delay, Sensory Processing Disorders (SPD), and ADHD & Others. It is complemented with the Gender Distribution Widget that reports the ratio of males to females while the clinical note emphasizes the higher prevalence rate of ASD among males at the 4:1 ratio.

The lower tables allocate the operational welfare distribution with respect to risk factors within households. The Center Utilization Distribution Table tracks utilization by the partner network, registering a sum of RM 18,500 in total subsidy distribution for 10 Active Centers, with specific percentages of patients per facility. These operational distributions are countered by the Household Income Bracket chart, which shows that 85% of all households fall under the B40 classification, and the Household Vulnerability Factors Matrix.

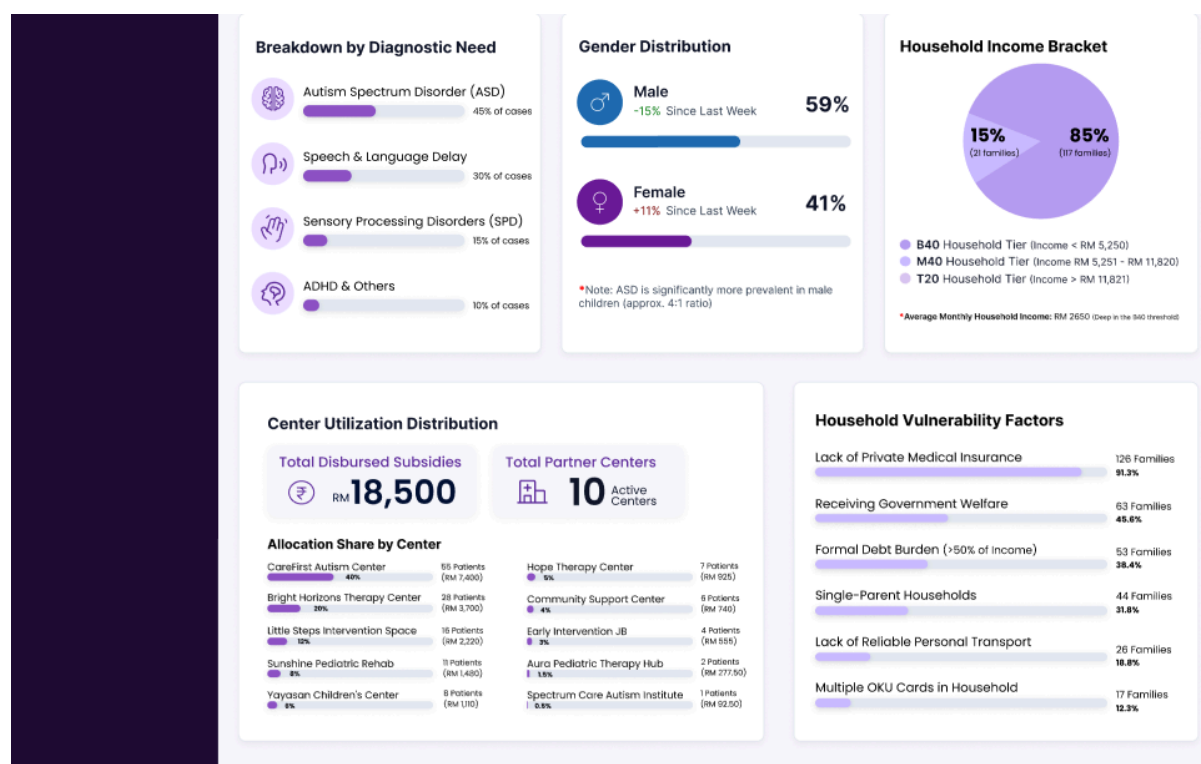


Figure 8.2.2 Regional Diagnostic Demographics and Household Socio-Economic Vulnerability Matrix

8.3 Figma Link

8.3.1 Parent

<https://www.figma.com/proto/wShhX78Mpja8IVeSgqrA18/SAD-Project?node-id=90-54&p=f&t=LYSQaHV6TnQseWTP-1&scaling=scale-down&content-scaling=fixed&page-id=70%3A54&starting-point-node-id=90%3A54&show-proto-sidebar=1>

8.3.2 NGO / Admin

<https://www.figma.com/proto/wShhX78Mpja8IVeSgqrA18/SAD-Project?node-id=1121-4222&p=f&t=ihXLcLCqvFxaYmcE-1&scaling=min-zoom&content-scaling=fixed&page-id=1121%3A4221>

9.0 SUMMARY OF THE PROPOSED SYSTEM

The suggested Autism Early Screening Support System aims at updating, digitizing and centralizing the tracking process of home care and consultations for parents dealing with early signs of autism and development delay in their children. In this regard, the existing paper-based process is based mainly on disconnected paper documentation and disconnected physical flashcards which leads to confusion regarding the evaluation, increased cognitive load for the caregiver, and lack of continuity after treatment. In order to solve this problem, the suggested system provides a digital solution designed to facilitate early data collection, automated trends analysis, and seamless connection with screening services.

9.1 Core Functional Modules

The architecture of the proposed system optimizes user autonomy, early identification, and institutional synchronization through five specialized core functions:

- **Behaviour Monitoring:** To ensure continuous tracking and databased insights following clinical intervention, this module provides parents with a structured framework to log routine behaviors and critical incidents. By translating daily observation data into clinical insights, the module helps identify early developmental regressions or emergent behavioral trends.

The monitoring functionality is split between two main logging processes:

1. **Behavior Observation Logging:** Parents log routine clinical parameters in four major areas of development: Social Interaction, Communication, Sensory Sensitivity and Repetitive and Restricted Behaviors. The information is summarized to produce the Behavioral Pattern Report, which focuses on common triggers and irregularities for each week period. All the structured logs can be saved, analyzed retrospectively and exported as a clinical PDF report.
 2. **Meltdown & Tantrum Incident Logging:** Designed specifically for severe behavioral occurrences, this process lets parents log acute behavioral episodes in a structured manner. The logging process collects relevant contextual information such as date and time, length of episode, triggers (for example, sensory over-stimulation or change in schedule), levels of intensity, and precise reaction. For greater accuracy of data, parents may include relevant media files (pictures/video) as objective clinical evidence along with timelines.
- **Center Booking Appointment:** In order to fill the post-discharge gap, the module features a real-time scheduling engine. In case the system detects any alarming behavioral patterns or serious meltdowns, parents can immediately bypass the paperwork and schedule consultation sessions directly with centers that have been verified.

For flexible booking depending on urgency levels, the module offers two separate booking channels:

1. **Pre-assigned Specialist Model:** Suitable for therapeutic continuity where parents can book sessions with specific and dedicated specialists according to long-term treatment plans.
 2. **Dynamic Assignment on Arrival:** Suitable for quick triaging, first-time visits, or parents who cannot decide on the particular specialists. In this method, families can schedule the appointment slot first. On arriving at the center, the professional team assesses the needs of the child and assigns them to the appropriate specialist.
- **Help Guideline:** In order to maintain a constant stream of user onboarding and technical self-sufficiency, this module utilizes the use of a centralized support knowledge base and triage hub. Created to minimize administrative chokepoints, the Help Center gives parents the ability to troubleshoot problems directly, through instructional resources, and communication avenues.

This support model has been constructed through three key touchpoints:

1. **Contact Support Gateways:** Provides multiple layers of escalation options for immediate and non-immediate help. These include an instant messaging tool showing available agents, an email address for support that receives replies within 24 hours, and a toll-free phone number.
 2. **Quick Guides:** A selected archive of procedural guides that focus on important user processes. This is where parents will be guided to conduct M-CHAT-R screeners, fill out organizational profiles (like the BPPOKU Bahagian A), understand risk classifications, and set up the behavior tracking modules.
 3. **Frequently Asked Questions (FAQs):** An expandable accordion-based FAQ engine that provides answers to commonly asked questions about platform processes, screening frequency, admin permissions, and strong privacy controls for users.
- **Special Educational Center:** The current module consists of a curated and verified regional directory of specialized autism treatment facilities and schools. For facilitating localized placements, this module acts as an intelligent directory for nearby specialized facilities. The module uses the user's geolocation to assist parents in evaluating and filtering out appropriate special education systems.

The intelligent directory is structured into two pathways of institutions based on different family requirements:

1. **Private Centres:** Features private institutions near the user with their unique special programs and flexibility in schedule. Each card consists of relevant information, including distance parameters, user reviews, contact registry, and badges indicating their specific therapy (e.g., Autism Therapy, Speech Therapy).

2. **Public Centres:** Consists of government-sponsored and highly subsidized programs like schools integration programs (SMK, SJK). This helps lower income families to locate their nearby public facilities without spending much money.
- **Community Groups Module:** In order to counter caregiver loneliness and build peer camaraderie, this module acts as a local community that links up caregivers to their respective communities. Users are able to browse through a range of communities from various categories, such as Diagnosis Support, Sensory Help, School/Education, and Local Get-Togethers. The community cards keep track of active member databases and daily posting stats, where parents can immediately join these communities and keep up to date with real time discussions and peer chat windows.

9.2 Technical, Operational, and Economic Impact

Technical Feasibility: The design utilizes cloud-native synchronization that makes it possible to match calendars on the spot for both booking systems and process behavioral diaries automatically. Moreover, the highly reusable UI components used in duplicate booking flows reduce unnecessary code bloat, whereas asynchronous queuing allows parents to keep a track record of their child's daily behaviors and meltdowns off-line, even using low-specced devices without exhausting their memory capacity.

Operational Efficiency: The design provides a visually attractive but definitely clutter-free interface that greatly reduces the learning curve for parents and therapy center staff in their hectic schedules. With digitalization, on-site clinical teams will be able to identify the needs of children right away through exported PDF behavioral reports, as well as minimize time-consuming admin activities and support tickets via Help Center quick guides.

Economic Value: Although the initial launch of the system entails some development costs, the system is very economically viable through the optimization of clinical matching and the avoidance of physical tracking costs. It preserves the funds of NGOs through the dynamic refunding of any unused budgets of sessions back to the fund pool, and also subsidization of centers according to vulnerability criteria.

9.3 Conclusion

The Autism Early Screening Support System that is being advocated provides a strong and future-proof IT system that has been designed to address the weak points of the traditional paper-based systems of home monitoring. The new system provides a major advancement in the area by providing a more efficient way of recording information at home through an interactive button system behavioral diary that converts subjective observations to a graphical timeline format. This not only takes away all chances of misinterpretation but also provides a direct digital access for professional interventions through its scheduling system.

Nevertheless, a number of operational constraints exist in that the software is still highly reliant on the cooperation of the caregivers to regularly input data manually into the Behavior

Monitoring component. Furthermore, its real time appointment matching feature requires constant high speed Internet connectivity while the help information and the school directory will need to be constantly curated to avoid any obsolescence in the data. Despite all these constraints, the software offers an important value addition to local health informatics in the form of filling the gap of "post-therapy isolation" commonly experienced by the family after discharge from clinical care.

In order to optimize the efficiency and functionality of the system in terms of the longevity of both the technology and the utility, there are certain improvements that could be implemented in future works. For instance, by incorporating predictive artificial intelligence algorithms, the system would use machine learning algorithms to analyze the previous behaviors from the diary and pinpoint any hidden environmental factors that could trigger meltdowns or overstimulation. Additionally, by improving the incident logging module to include computer vision algorithms, the system would automatically analyze the video clips uploaded by the parents and detect any specific repetitive physical movement patterns such as hand-flapping. Lastly, incorporating the Special Educational Center module as a dynamic environment would enable partner schools to conduct IEP and distance learning activities in the system.